AVAZBEKOV SALOHIDDIN

SAT 1200+

O'ZBEKISTON DAVLAT UNIVERSITETLARI UCHUN TAYYORLOV

SAT *math* 0-1

FOUNDATION BOOK

QISQA MUDDAT ICHIDA SAT
UCHUN KERAKLI MATEMATIKA
VA IZGLIZ TILI BILIMLARI

AVAZBEKOV SALOHIDDIN AZAMJON O’G’LI

SAT 1200+

1-KITOB · ASOSLAR

O’zbekiston davlat universitetlari uchun tayyorlov

Qisqa muddat ichida SAT uchun kerakli matematika
va ingliz tili bilimlari

Mualliflik huquqi

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Kitob haqida

SAT 1200+ — Digital SAT imtihonining matematika bo'limiga tayyorlanish uchun yozilgan uch kitoblik seriya. Har bir kitob talabani asta-sekin yuqori ball tomon olib boradi: **1-kitob** asosiy matematik fundamentni mustahkamlaydi, **2-kitob** SATning asosiy mavzularini (algebra, funksiyalar, statistika, geometriya) to'liq ingliz tilida yoritadi, **3-kitob** esa yuqori ballga erishish uchun murakkab masalalar va tezlik strategiyalarini o'rgatadi.

Ushbu kitob zamonaviy, ma'lumotlarga asoslangan metodika asosida tuzilgan: har bir mavzu College Board'ning rasmiy savollar bazasi tahlili, O'zbekiston maktab dasturi bilan moslik va xalqaro ta'lim tizimlaridagi eng yaxshi amaliyotlarni hisobga olgan holda tanlangan.

Har bir darsning tuzilishi:

1. **Qoida/Formula sahifasi** — mavzuning eng muhim qoidalari bitta sahifada, aniq misollar bilan.
2. **Tushuncha (Concept)** — har bir mavzu ta'rif, yechilgan misol va mashq bilan tushuntiriladi.
3. **Lug'at (Vocabulary)** — SAT savollarida uchraydigan ingliz tilidagi atamalar o'zbekcha tarjimasi bilan.
4. **Mashqlar (Problems)** — bosqichma-bosqich qiyinlashib boruvchi mashqlar to'plami.

Kitob oxirida esa barcha 9 ta darsda uchragan so'zlarni o'z ichiga olgan **to'liq lug'at (complete glossary)** berilgan — bu 2-kitobga o'tishda talaba allaqachon tanish bo'lgan ingliz tilidagi SAT lug'ati bilan qurollangan bo'ladi.

1 Asosiy arifmetika va amallar tartibi

FORMULA SHEET

QOIDA · Amal nomlari

Belgi	Amal	Natija nomi	English
$+$	Qo'shish	Yig'indi	Sum
$-$	Ayirish	Ayirma	Difference
$\times$ / ab	Ko'paytirish	Ko'paytma	Product
$\frac{a}{b}$	Bo'lish	Bo'linma	Quotient

QOIDA · Bajarish tartibi

Har bir amal o'z bosqichiga ega. Bitta misolda barcha to'rt bosqich ketma-ket qanday ishlashini ko'ring:

1. Qavs | **2. Daraja / Ildiz** | **3. × ÷** | **4. + −**

$$5 + 2 \times \underline{(8-3)}^{2} \div 5 \quad \leftarrow \text{1. qavs}$$
$$\downarrow$$
$$5 + 2 \times \underline{5^{2}} \div 5 \quad \leftarrow \text{2. daraja}$$
$$\downarrow$$
$$5 + \underline{2 \times 25 \div 5} \quad \leftarrow \text{3. ×÷ (chapdan o'ngga)}$$
$$\downarrow$$
$$\underline{5 + 10} \quad \leftarrow \text{4. + −}$$
$$\downarrow$$
$$= 15$$

1-rasm. Bitta ifodaning to'rtala bosqich bo'yicha ketma-ket soddalashuvi.

QOIDA · Qavs ochish qoidalari

Qoida	Misol
$-(+a) = -a$	$-(+5) = -5$
$-(-a) = +a$	$-(-53) = +53$
$+(-a) = -a$	$+(-22) = -22$
$+(+a) = +a$	$+(+52) = +52$

1 Asosiy arifmetika va amallar tartibi

VOCABULARY – LUG'AT

Belgi	English	O'zbekcha
+	sum	*yig'indi*
−	difference	*ayirma*
×	product	*ko'paytma*
÷	quotient	*bo'linma*
x^2	exponent	*daraja*
x^2	exponents	*daraja*
=	equal	*teng*
$ax + b = c$	equation	*tenglama*
$ax + b = c$	equations	*tenglamalar*
$3x + 2$	expression	*ifoda*
$3x + 2$	expressions	*ifodalar*
–	value	*qiymat*
–	values	*qiymatlar*
+	positive	*musbat*
−	negative	*manfiy*
>	greater	*katta*
–	greatest	*eng katta*
–	error	*xato*
–	errors	*xatolar*
–	following	*quyidagi*
∴	therefore	*demak*
1 → 2 → 3 → 4	order	*tartib*
( )	parentheses	*qavs*
ℤ	integer	*butun son*
ℤ	integers	*butun sonlar*
0	zero	*nol*
−	minus	*minus*
+	plus	*plyus*
–	result	*natija*
Σ	total	*jami*

1-to'plam. *To'rt bosqichli odatni mustahkamlang: qavs → daraja → II bosqich → I bosqich.*

1. $3 + 4 \times 2 =$

2. $(3 + 4) \times 2 =$

3. $20 - 6 \div 3 =$

4. $(18 - 6) \div 3 =$

5. $2 \times 3^2 =$

6. $(2 \times 3)^2 =$

7. $18 \div 3 \times 2 =$

8. $10 - 2 \times 3 + 4 =$

2-to'plam. *Ishorali sonlar berilgan qiymat sifatida – tartibni qo'llashga e'tibor bering.*

1. $-3 + 4 \times 2 =$

2. $(-3 + 4) \times 2 =$

3. $5 - (-2) \times 3 =$

4. $(-2)^2 \times 3 =$

5. $-2^2 \times 3 =$

6. $10 + (-4) \div 2 =$

7. $(10 + (-4)) \div 2 =$

8. $-5 \times (-3) + 2 =$

3-to'plam. *Kasr chizig'ini guruhlash belgisi sifatida qo'llang.*

1. $\dfrac{3+5}{2\times 2} =$

2. $\dfrac{12-4}{2\times 2} =$

3. $\dfrac{2+3\times 4}{7} =$

4. $\dfrac{5^2-1}{3\times 2} =$

5. $\dfrac{6+2}{2+2} =$

6. $\dfrac{20}{2+3} - 1 =$

7. $\dfrac{18-6}{3\times(1+1)} =$

8. $\dfrac{7+9}{4^2-12} =$

4-to'plam. *Modul belgisini guruhlash belgisi sifatida qo'llang.*

1. $|5-9| =$

2. $|5-9|+3 =$

3. $-|5-9| =$

4. $|3-7|\times 2 =$

5. $|-3\times 4| =$

6. $|6-10|-|2-5| =$

7. $10-|4-12| =$

8. $|(-2)^2-9| =$

5-to’plam. *Aralash – haqiqiy test zichligi: daraja, kasr chizig’i va qavs birgalikda.*

1. $\dfrac{2^3+4}{6-4}-1=$

2. $5+\dfrac{(6-2)^2}{4}=$

3. $\dfrac{10-2\times 3}{2^2}=$

4. $3\times(4+2^2)-\dfrac{10}{2}=$

5. $\dfrac{5^2-3\times 5}{2+3}=$

6. $2+3\times(7-2^2)=$

7. $\dfrac{8+|3-7|}{2\times 2}=$

8. $(-3)^2+2\times(5-8)=$

6-to’plam. *Xatoni toping – buzilgan qadamni ayting, to’g’ri javobni bering.*

1. $20\div 4\times 5$ ifodasida ko’rsatilgan yechimda xato bor:
   “1-qadam: $4\times 5=20$. *2-qadam:* $20\div 20=1$.”
   Xato va to’g’ri javob: .....

2. $15-3+4$ ifodasida ko’rsatilgan yechimda xato bor:
   “1-qadam: $3+4=7$. *2-qadam:* $15-7=8$.”
   Xato va to’g’ri javob: .....

3. $\dfrac{10+2}{2\times 3}$ ifodasida ko’rsatilgan yechimda xato bor:
   “Avval $2\times 3=6$, *so’ng* $2\div 6=\frac{1}{3}$, *so’ng* $10+\frac{1}{3}$.”
   Xato va to’g’ri javob: .....

4. -4^2 ifodasida ko’rsatilgan yechimda xato bor:
   “$-4^2=(-4)^2=16$.”
   Xato va to’g’ri javob: .....

5. $3-|5-9|$ ifodasida ko’rsatilgan yechimda xato bor:
   “$3-5-9=-11$.”
   Xato va to’g’ri javob: .....

7-to’plam. *Bonus / qiyin – har biriga taxminan 90 soniya.*

1. $|(-3)^2 - 2 \times (5 - 8)| \div 3 =$

2. $\dfrac{4 + |2 - 6|^2}{3 \times 2 - 2} =$

3. $2 \times \left[(10 - 4)^2 \div 3 + 1\right] - 5 =$

4. $\dfrac{|-12| + 3^2}{7 - 4} =$

5. $1 + 2 \times \left[|4 - 10| - (3 - 1)^2\right] =$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	11	2	14
	3	18	4	4
	5	18	6	36
	7	12	8	8
2-to'plam	1	5	2	2
	3	11	4	12
	5	-12	6	8
	7	3	8	17
3-to'plam	1	2	2	2
	3	2	4	4
	5	2	6	3
	7	2	8	4
4-to'plam	1	4	2	7
	3	-4	4	8
	5	12	6	1
	7	2	8	5
5-to'plam	1	5	2	9
	3	1	4	19
	5	2	6	11
	7	3	8	3
7-to'plam	1	5	2	5
	3	21	4	7
	5	5	–	–
6-to'plam	1. Xato 1-qadamda – bo'lish chapda, lekin ko'paytirish avval bajarilgan. To'g'ri: $20 \div 4 = 5$, so'ng $5 \times 5 = 25$.			
	2. Xato 1-qadamda – ayirish chapda, lekin qo'shish avval bajarilgan. To'g'ri: $15-3 = 12$, so'ng $12+4 = 16$.			
	3. Kasr chizig'i suratni ham guruhlaydi. To'g'ri: $\frac{12}{6} = 2$.			
	4. Qavs yo'q – daraja faqat 4 ga tegishli. To'g'ri: $-4^2 = -16$.			
	5. Modul ichidagi amal avval bajariladi. To'g'ri: $5 - 9 = -4$, $\lvert -4 \rvert = 4$, $3 - 4 = -1$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram

FORMULA SHEET

QOIDA · Surat, maxraj, soddalashtirish

surat → $\frac{24}{36}$ ← maxraj

$\xrightarrow{\div 12}$ $\frac{2}{3}$

eng oddiy ko'rinish

QOIDA · Qo'shish / ayirish

Maxrajlar har xil bo'lsa, avval umumiy maxrajga keltiring – faqat shundan keyin suratlarni qo'shing yoki ayiring:

$\frac{1}{4} + \frac{1}{6}$ ← 1. EKUK(4,6)=12

↓

$\frac{3}{12} + \frac{2}{12}$ ← 2. suratlarni qo'shing

↓

$= \frac{5}{12}$

QOIDA · Ko'paytirish

$$\frac{4}{9} \times \frac{3}{8}$$

qisqartiring: 4&8 → ÷4, 3&9 → ÷3

$$= \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$$

QOIDA · Bo'lish

$$\frac{3}{4} \div \frac{2}{5}$$

teskarisiga ko'paytiring (K-C-F)

$$= \frac{3}{4} \times \frac{5}{2} = \frac{15}{8}$$

QOIDA · Aralash son ↔ noto'g'ri kasr

$3\frac{2}{5}$ $\xrightarrow{3\times5+2}$ $\frac{17}{5}$ $\qquad$ $\frac{23}{6}$ $\xrightarrow{23\div6}$ $3\frac{5}{6}$

Butun son maxrajga ko'paytiriladi, suratga qo'shiladi – maxraj o'zgarmaydi. Teskarisida: suratni maxrajga bo'ling, bo'linma – butun son, qoldiq – yangi surat.

VOCABULARY – LUG'AT

Belgi	English	O'zbekcha
$\frac{a}{b}$	fraction	*kasr*
–	numerator / denominator	*surat / maxraj*
–	simplify / reduce	*soddalashtirish / qisqartirish*
–	common denominator	*umumiy maxraj*
–	LCM (least common multiple)	*EKUK (eng kichik umumiy karrali)*
–	GCD (greatest common divisor)	*EKUB (eng katta umumiy bo'luvchi)*
$\frac{b}{a}$	reciprocal	*teskari son*
$2\frac{1}{2}$	mixed number	*aralash son*
$\frac{17}{5}$	improper fraction	*noto'g'ri kasr*
$\times$	multiply	*ko'paytirish*
$\div$	divide	*bo'lish*
$=$	equal	*teng*
$=$	equivalent	*teng kuchli*
–	whole	*butun*
$+$	sum	*yig'indi*
$-$	difference	*ayirma*
$\times$	product	*ko'paytma*
$\div$	quotient	*bo'linma*
$>$	greater	*katta*
–	value	*qiymat*
–	total	*jami*
$\frac{1}{2}$	half	*yarim*
–	remaining	*qolgan*
–	result	*natija*
–	common	*umumiy*

1-to'plam. *Qisqartiring – javobni eng sodda ko'rinishda yozing.*

1. $\frac{8}{12} =$
2. $\frac{15}{25} =$
3. $\frac{18}{24} =$
4. $\frac{21}{14} =$
5. $\frac{36}{48} =$
6. $\frac{50}{75} =$
7. $\frac{28}{35} =$
8. $\frac{100}{40} =$

2-to'plam. *Qo'shing yoki ayiring – avval umumiy maxraj toping.*

1. $\frac{1}{3} + \frac{1}{4} =$
2. $\frac{2}{5} + \frac{1}{3} =$
3. $\frac{3}{4} - \frac{1}{6} =$
4. $\frac{5}{6} - \frac{2}{9} =$
5. $\frac{1}{2} + \frac{2}{3} - \frac{1}{4} =$
6. $\frac{7}{10} - \frac{1}{4} =$
7. $\frac{3}{8} + \frac{5}{12} =$
8. $\frac{11}{12} - \frac{3}{4} =$

3-to'plam. *Ko'paytiring – hisoblashdan oldin qisqartirishga harakat qiling.*

1. $\frac{2}{3} \times \frac{3}{5} =$
2. $\frac{4}{7} \times \frac{7}{8} =$
3. $\frac{5}{6} \times \frac{2}{3} =$
4. $\frac{3}{4} \times \frac{8}{9} =$
5. $\frac{7}{10} \times \frac{5}{14} =$
6. $\frac{2}{9} \times \frac{3}{4} =$
7. $\frac{5}{8} \times \frac{4}{15} =$
8. $\frac{9}{10} \times \frac{5}{6} =$

4-to'plam. *Bo'ling – ikkinchi kasrni teskarisiga aylantirib ko'paytiring.*

1. $\frac{2}{3} \div \frac{4}{5} =$
2. $\frac{3}{4} \div \frac{1}{2} =$
3. $\frac{5}{6} \div \frac{5}{9} =$
4. $\frac{7}{8} \div \frac{7}{4} =$
5. $\frac{2}{5} \div \frac{3}{10} =$
6. $\frac{9}{10} \div \frac{3}{5} =$
7. $\frac{4}{9} \div \frac{8}{3} =$
8. $\frac{5}{12} \div \frac{5}{6} =$

5-to'plam. *Aralash son ↔ noto'g'ri kasr.*

1 $2\frac{3}{4}$ ni noto'g'ri kasrga aylantiring.

2 $5\frac{1}{6}$ ni noto'g'ri kasrga aylantiring.

3 $\frac{17}{5}$ ni aralash songa aylantiring.

4 $\frac{29}{4}$ ni aralash songa aylantiring.

5 $1\frac{1}{2} + 2\frac{1}{3} =$ (aralash son)

6 $3\frac{1}{4} - 1\frac{3}{4} =$ (aralash son)

7 $2\frac{2}{5} \times 1\frac{1}{4} =$

8 $\frac{53}{6}$ ni aralash songa aylantiring.

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1 $\frac{1}{3} + \frac{1}{4}$ ifodasida ko'rsatilgan yechimda xato bor:

"$\frac{1+1}{3+4} = \frac{2}{7}$."

Xato va to'g'ri javob: .....

2 $\frac{2}{5} + \frac{2}{5}$ ifodasida ko'rsatilgan yechimda xato bor:

"Suratni saqlab, maxrajlarni qo'shamiz: $\frac{2}{10}$*."*

Xato va to'g'ri javob: .....

3 $\frac{3}{4} \div \frac{2}{5}$ ifodasida ko'rsatilgan yechimda xato bor:

"$\frac{4}{3} \times \frac{2}{5} = \frac{8}{15}$."

Xato va to'g'ri javob: .....

4 $\frac{16}{24}$ ni soddalashtirishda ko'rsatilgan yechimda xato bor:

"2 ga qisqartirib, $\frac{8}{12}$ *– tayyor."*

Xato va to'g'ri javob: .....

5 $2\frac{3}{4}$ ni noto'g'ri kasrga aylantirishda ko'rsatilgan yechimda xato bor:

"$\frac{2+3}{4} = \frac{5}{4}$."

Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – har biriga taxminan 90 soniya.*

1. $\left(\frac{1}{2}+\frac{1}{3}\right)\times\frac{2}{5}=$

2. $3\frac{1}{2}\div\left(\frac{1}{4}+\frac{1}{4}\right)=$

3. $\left(\frac{2}{3}-\frac{1}{6}\right)\times\left(\frac{3}{4}+\frac{1}{2}\right)=$

4. $1\frac{3}{4}\times\frac{2}{7}+\frac{1}{8}=$

5. $\left(\frac{5}{6}\div\frac{5}{9}\right)-\left(\frac{1}{2}\times\frac{2}{3}\right)=$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	$\frac{2}{3}$	2	$\frac{3}{5}$
	3	$\frac{3}{4}$	4	$\frac{3}{2}$
	5	$\frac{3}{4}$	6	$\frac{2}{3}$
	7	$\frac{4}{5}$	8	$\frac{5}{2}$
2-to'plam	1	$\frac{7}{12}$	2	$\frac{11}{15}$
	3	$\frac{7}{12}$	4	$\frac{11}{18}$
	5	$\frac{11}{12}$	6	$\frac{9}{20}$
	7	$\frac{19}{24}$	8	$\frac{1}{6}$
3-to'plam	1	$\frac{2}{5}$	2	$\frac{1}{2}$
	3	$\frac{5}{9}$	4	$\frac{2}{3}$
	5	$\frac{1}{4}$	6	$\frac{1}{6}$
	7	$\frac{1}{6}$	8	$\frac{3}{4}$
4-to'plam	1	$\frac{5}{6}$	2	$\frac{3}{2}$
	3	$\frac{3}{2}$	4	$\frac{1}{2}$
	5	$\frac{4}{3}$	6	$\frac{3}{2}$
	7	$\frac{1}{6}$	8	$\frac{1}{2}$
5-to'plam	1	$\frac{11}{4}$	2	$\frac{31}{6}$
	3	$3\frac{2}{5}$	4	$7\frac{1}{4}$
	5	$3\frac{5}{6}$	6	$1\frac{1}{2}$
	7	3	8	$8\frac{5}{6}$
7-to'plam	1	$\frac{1}{3}$	2	7
	3	$\frac{5}{8}$	4	$\frac{5}{8}$
	5	$\frac{7}{6}$	–	–
6-to'plam	1. Surat/maxraj mustaqil qo'shilgan, umumiy maxraj topilmagan. To'g'ri: $\frac{1}{3} = \frac{4}{12}$, $\frac{1}{4} = \frac{3}{12}$, yig'indi $\frac{7}{12}$.			
	2. Maxrajlar teng bo'lsa, faqat suratlar qo'shiladi. To'g'ri: $\frac{2}{5} + \frac{2}{5} = \frac{4}{5}$.			
	3. Teskari kasr ikkinchi (bo'luvchi) kasrga qo'llanishi kerak edi. To'g'ri: $\frac{3}{4} \times \frac{5}{2} = \frac{15}{8}$.			
	4. Bitta qisqartirishdan keyin to'xtagan (8 va 12 hali 4 ga bo'linadi). To'g'ri: $\frac{16}{24} = \frac{2}{3}$.			
	5. Butun son maxrajga ko'paytirilmagan. To'g'ri: $3 \times 4 = 12$, $12 + 3 = 15$, ya'ni $\frac{11}{4}$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram

FORMULA SHEET

QOIDA · Nuqtani tekislash

$$\begin{array}{r} 3.70 \\ +2.45 \\ \hline 6.15 \end{array}$$

Qo’shish/ayirishdan oldin nuqtalarni ustma-ust tekislang; yetishmagan xonani nol bilan to’ldiring.

QOIDA · O’nli kasr ↔ oddiy kasr

$$0.75 = \frac{75}{100} = \frac{3}{4} \quad (\div 25,\ \text{EKUB}) \qquad \frac{1}{8} \xrightarrow{\div} 0.125$$

Oxirgi xona qiymati maxrajni belgilaydi; teskarisida suratni maxrajga bo’ling.

QOIDA · Foiz ↔ o’nli kasr

$$45\% \xrightarrow{\div 100} 0.45 \qquad 0.6 \xrightarrow{\times 100} 60\%$$

Foizdan o’nli kasrga: vergulni 2 xona chapga suring. Teskarisida: 2 xona o’ngga.

QOIDA · Sonning foizi

80 ning 25%i

↓ 1. 25% = 0.25

0.25×80

↓ 2. ko’paytiring

$= 20$

VOCABULARY – LUG’AT

Belgi	English	O’zbekcha
%	**percent**	*foiz*
%	**percentage**	*foiz*
%	**percentages**	*foizlar*
–	**decimal**	*o’nlik*
$\frac{a}{b}$	**fraction**	*kasr*
$\frac{a}{b}$ (top)	**numerator**	*surat*
$\frac{a}{b}$ (bottom)	**denominator**	*maxraj*
=	**equivalent**	*teng kuchli*
=	**equal**	*teng*
–	**value**	*qiymat*
–	**values**	*qiymatlar*
–	**whole**	*butun*
>	**greater**	*katta*
≈	**approximately**	*taxminan*
–	**nearest**	*eng yaqin*
–	**round**	*yaxlitlamoq*
×	**multiplying**	*ko’paytirish*
×	**multiplied**	*ko’paytirildi*
÷	**dividing**	*bo’lish*
÷	**divided**	*bo’lingan*
–	**difference**	*ayirma*
+	**sum**	*yig’indi*
–	**subtracting**	*ayirish*
+	**adding**	*qo’shish*
–	**number**	*son*
–	**numbers**	*sonlar*
–	**total**	*jami*
–	**simplifying**	*soddalashtirish*
.	**decimal point**	*vergul (sat-da: nuqta)*
–	**place value**	*xona qiymati*
–	**to align**	*tekislamoq*
–	**to convert**	*aylantirmoq*
gcd	**greatest common divisor**	*eng katta umumiy bo’luvchi*
–	**percent of a number**	*sonning foizi*

1-to’plam. *O’nli kasrlarni qo’shish va ayirish: vergulni tekislashni unutmang.*

1. $3.7 + 2.45 =$

2. $12.6 - 4.75 =$

3. $0.8 + 0.09 =$

4. $15 - 3.25 =$

5. $2.05 + 1.95 =$

6. $100 - 0.4 =$

7. $0.125 + 0.875 =$

8. $7.3 - 2.08 =$

2-to’plam. *O’nli kasrlarni ko’paytirish va bo’lish – kasr xonalarini to’g’ri sanang.*

1. $0.3 \times 0.4 =$

2. $2.5 \times 1.2 =$

3. $0.06 \times 5 =$

4. $1.44 \div 1.2 =$

5. $0.5 \times 0.5 =$

6. $7.2 \div 0.4 =$

7. $0.02 \times 0.03 =$

8. $9.6 \div 2.4 =$

3-to’plam. *O’nli kasr ↔ oddiy kasr almashtirish.*

1. 0.5 ni oddiy kasrga aylantiring.

2. 0.25 ni oddiy kasrga aylantiring.

3. 0.75 ni oddiy kasrga aylantiring.

4. 0.2 ni oddiy kasrga aylantiring.

5. $\frac{3}{4}$ ni o’nli kasrga aylantiring.

6. $\frac{1}{8}$ ni o’nli kasrga aylantiring.

7. $\frac{7}{10}$ ni o’nli kasrga aylantiring.

8. 0.125 ni oddiy kasrga aylantiring.

4-to’plam. *Foiz ↔ o’nli kasr almashtirish – 100% dan katta va 1% dan kichik holatlarni ham o’z ichiga oladi.*

1. 45% ni o’nli kasrga aylantiring.

2. 7% ni o’nli kasrga aylantiring.

3. 150% ni o’nli kasrga aylantiring.

4. 0.5% ni o’nli kasrga aylantiring.

5. 0.6 ni foizga aylantiring.

6. 1.25 ni foizga aylantiring.

7. 0.03 ni foizga aylantiring.

8. 2 ni foizga aylantiring.

5-to'plam. *Foiz ↔ oddiy kasr, va sonning foizini topish.*

1. 80 ning 25% i qanchaga teng?

2. 150 ning 10% i qanchaga teng?

3. 45 ning 60% i qanchaga teng?

4. $\frac{1}{4}$ ni foizga aylantiring.

5. $\frac{3}{5}$ ni foizga aylantiring.

6. 150% ni oddiy kasrga aylantiring.

7. 50 ning 12% i qanchaga teng?

8. 20 ning 90% i qanchaga teng?

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1. 80 ning 45% ini hisoblashda ko'rsatilgan yechimda xato bor:

   "$45 \times 80 = 3600$, *demak javob* 3600."

   Xato va to'g'ri javob: .....

2. $3.7 + 2.45$ ni hisoblashda ko'rsatilgan yechimda xato bor:

   "Vergulni tekislamasdan qo'shib, $3.7 + 2.45 = 2.82$ *deb topilgan."*

   Xato va to'g'ri javob: .....

3. $1.44 \div 1.2$ ni hisoblashda ko'rsatilgan yechimda xato bor:

   "Kasr xonalarini noto'g'ri sanab, $1.44 \div 1.2 = 0.12$ *deb topilgan."*

   Xato va to'g'ri javob: .....

4. 0.2 ni oddiy kasrga aylantirishda ko'rsatilgan yechimda xato bor:

   "0.2 *dagi raqam yuzdan bir xonasida deb o'ylab,* $\frac{2}{100} = \frac{1}{50}$ *deb topilgan."*

   Xato va to'g'ri javob: .....

5. 150% ni o'nli kasrga aylantirishda ko'rsatilgan yechimda xato bor:

   "Vergul faqat bir xona chapga surilib, $150\% = 15$ *deb topilgan."*

   Xato va to'g'ri javob: .....

7-to’plam. *Bonus / qiyin – aralash almashtirishlar, har biriga taxminan 90 soniya.*

1 0.8 ning 35% ini toping, so’ngra natijani 100 ga ko’paytiring.

2 $\frac{3}{8}$ ni foizga aylantiring.

3 0.6 ning necha foizi 0.3 ga teng?

4 1.5 ni foizga va oddiy kasrga aylantiring.

5 (0.25 + 0.5) yig’indisining 40% ini toping.

Javoblar

To’plam	#	Javob	#	Javob
1-to’plam	1	6.15	2	7.85
	3	0.89	4	11.75
	5	4	6	99.6
	7	1	8	5.22
2-to’plam	1	0.12	2	3
	3	0.3	4	1.2
	5	0.25	6	18
	7	0.0006	8	4
3-to’plam	1	1/2	2	1/4
	3	3/4	4	1/5
	5	0.75	6	0.125
	7	0.7	8	1/8
4-to’plam	1	0.45	2	0.07
	3	1.5	4	0.005
	5	60%	6	125%
	7	3%	8	200%
5-to’plam	1	20	2	15
	3	27	4	25%
	5	60%	6	3/2
	7	6	8	18
7-to’plam	1	28	2	37.5%
	3	50%	4	150%, 3/2
	5	0.3	–	–
6-to’plam	1. Foiz to’g’ridan-to’g’ri ko’paytirilgan, o’nli kasrga aylantirilmagan. To’g’ri: $45\% = 0.45$, so’ng $0.45 \times 80 = 36$.			
	2. Vergullar tekislanmagan. To’g’ri: $3.70 + 2.45 = 6.15$.			
	3. Kasr xonalari noto’g’ri sanalgan. To’g’ri: $1.44 \div 1.2 = 1.2$.			
	4. Raqamning xonasi noto’g’ri aniqlangan (yuzdan emas, o’ndan). To’g’ri: $0.2 = \frac{2}{10} = \frac{1}{5}$.			
	5. Vergul bir xona o’rniga ikki xona chapga surilishi kerak edi. To’g’ri: $150\% = 1.50 = 1.5$.			

Video yechimlar va qo’shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram

FORMULA SHEET

QOIDA · Musbat/manfiy va modul

Son	Modul	Qarama-qarshi son
7	$\|7\| = 7$	-7
-7	$\|-7\| = 7$	7
-12	$\|-12\| = 12$	12

QOIDA · Qo'shish / ayirish

Bir xil ishora: modullarni qo'shing, ishorani saqlang. Har xil ishora: kattasidan kichigini ayiring, kattaning ishorasini oling. Manfiyni ayirish – musbatini qo'shish: $a - (-b) = a + b$.

$(-7) + 12 - (-5)$ ← 1. $-(-5) = +5$

↓

$(-7) + 12 + 5$ ← 2. chapdan o'ngga

↓

$= 10$

QOIDA · Ko'paytirish / bo'lish

Ishoralar	Natija
bir xil	+
har xil	–

QOIDA · Necha marta manfiy?

$(-4) \times (-3) \div (-2)$

$(-4) \times (-3) = 12$ (bir xil)

$12 \div (-2) = -6$ (har xil)

$= -6$

QOIDA · Qavs ochish

$-(a + b) = -a - b$ $\quad -(a - b) = -a + b$

$-(7 - 4) = -7 + 4$

$-(7 - 4) + 3$

↓

$-7 + 4 + 3$ $\quad = 0$

VOCABULARY – LUG'AT

Belgi	English	O'zbekcha
+	**positive**	*musbat*
–	**negative**	*manfiy*
–	**opposite**	*qarama-qarshi*
–	**opposite numbers**	*qarama-qarshi sonlar*
$\|x\|$	**absolute value**	*modul*
–	**sign**	*ishora*
–	**same sign**	*bir xil ishora*
–	**different sign**	*har xil ishora*
0	**zero**	*nol*
–	**number line**	*sonlar o'qi*
=	**equal**	*teng*
>	**greater**	*katta*
–	**value**	*qiymat*
–	**values**	*qiymatlar*
$\mathbb{Z}$	**integer**	*butun son*
–	**integers**	*butun sonlar*
–	**term**	*had*
–	**terms**	*hadlar*
–	**even**	*juft*
–	**odd**	*toq*
–	**result**	*natija*
–	**results**	*natijalar*
×	**multiply**	*ko'paytirmoq*
–	**multiplying**	*ko'paytirish*
÷	**dividing**	*bo'lish*
–	**number**	*son*
–	**numbers**	*sonlar*
–	**distance**	*masofa*
–	**true**	*to'g'ri*
–	**false**	*noto'g'ri*

1-to'plam. *Bir xil ishorali sonlarni qo'shing.*

1. $(-3) + (-8) =$
2. $(-15) + (-9) =$
3. $7 + 12 =$
4. $(-20) + (-5) =$
5. $(-11) + (-14) =$
6. $25 + 16 =$
7. $(-30) + (-12) =$
8. $9 + 18 =$

2-to'plam. *Aralash ishoralar va manfiyni ayirish.*

1. $8 - (-3) =$
2. $(-5) - (-12) =$
3. $(-7) + 10 =$
4. $14 - 20 =$
5. $(-9) - 6 =$
6. $6 - (-6) =$
7. $(-18) + 11 =$
8. $(-4) - (-4) =$

3-to'plam. *Ko'paytirish va bo'lish (ikki ko'paytuvchi).*

1. $(-6) \times 7 =$

2. $(-9) \times (-8) =$

3. $45 \div (-5) =$

4. $(-56) \div (-7) =$

5. $12 \times (-11) =$

6. $(-100) \div 4 =$

7. $(-13) \times (-3) =$

8. $(-81) \div (-9) =$

4-to'plam. *Ko'p ko'paytuvchili zanjirlar – manfiylar sonini sanang.*

1. $(-2) \times (-3) \times (-4) =$

2. $(-5) \times 6 \times (-2) =$

3. $100 \div (-5) \div (-2) =$

4. $(-1) \times (-1) \times (-1) \times (-1) =$

5. $(-3) \times (-3) \times (-3) =$

6. $48 \div (-4) \times (-3) =$

7. $(-10) \times (-10) \div 5 =$

8. $(-2) \times 3 \times (-4) \times (-1) =$

5-to'plam. *Qavslarni oching.*

1. $-(9-4)+6=$

2. $-(5+3)-2=$

3. $10-(6-11)=$

4. $-(-7+2)+1=$

5. $-(8-8)-5=$

6. $3-(-4-6)=$

7. $-(12-20)-9=$

8. $-(-6-(-2))+4=$

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1. $8-(-3)$ ifodasida ko'rsatilgan yechimda xato bor:

   "Noto'g'ri yechim: $8-3=5$*."*

   Xato va to'g'ri javob: .....

2. $(-5)\times(-6)$ ifodasida ko'rsatilgan yechimda xato bor:

   "Noto'g'ri yechim: -30*."*

   Xato va to'g'ri javob: .....

3. $-(10-4)$ ifodasida ko'rsatilgan yechimda xato bor:

   "Noto'g'ri yechim: $-10-4=-14$*."*

   Xato va to'g'ri javob: .....

4. $(-2)\times(-2)\times(-2)$ ifodasida ko'rsatilgan yechimda xato bor:

   "Noto'g'ri yechim: barcha ko'paytuvchilar manfiy bo'lgani uchun javob musbat: 8*."*

   Xato va to'g'ri javob: .....

5. $(-15)\div(-3)$ ifodasida ko'rsatilgan yechimda xato bor:

   "Noto'g'ri yechim: -5*."*

   Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – har biriga taxminan 90 soniya.*

1. $-(-8+3)\times(-2)-5=$

2. $(-4)\times(-3)-(-7)+(-2)\times 5=$

3. $18\div(-3)+(-2)\times(-4)-6=$

4. $-(5-(-9))+(-14)\div 7=$

5. $(-6)\times[(-2)-3]+(-10)\div(-5)=$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	-11	2	-24
	3	19	4	-25
	5	-25	6	41
	7	-42	8	27
2-to'plam	1	11	2	7
	3	3	4	-6
	5	-15	6	12
	7	-7	8	0
3-to'plam	1	-42	2	72
	3	-9	4	8
	5	-132	6	-25
	7	39	8	9
4-to'plam	1	-24	2	60
	3	10	4	1
	5	-27	6	36
	7	20	8	-24
5-to'plam	1	1	2	-10
	3	15	4	6
	5	-5	6	13
	7	-1	8	8
7-to'plam	1	-15	2	9
	3	-4	4	-16
	5	32	–	–
6-to'plam	1. Minus-minus (manfiyni ayirish) qo'shishga aylantirilmadi. To'g'ri: $8-(-3)=8+3=11$.			
	2. Bir xil ishorali ko'paytirish musbat natija beradi. To'g'ri: $(-5)\times(-6)=30$.			
	3. Manfiy ishora qavs ichidagi faqat birinchi hadga qo'llanildi. To'g'ri: $-(10-4)=-10+4=-6$.			
	4. Uchta (toq sondagi) manfiy ko'paytuvchi bor – toq son manfiylar natijani manfiy qiladi. To'g'ri: $(-2)\times(-2)\times(-2)=-8$.			
	5. Bo'lishda ham bir xil ishora musbat natija beradi. To'g'ri: $(-15)\div(-3)=5$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

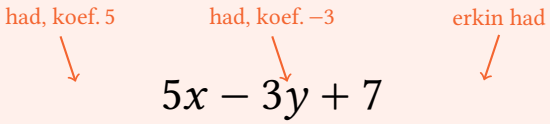
YouTube

QR kod

Telegram

FORMULA SHEET

QOIDA · Had, koeffitsient, o'xshash hadlar



$4x + 3y - x + 7$ —(x hadlarni qo'shing)→ $3x + 3y + 7$

QOIDA · Taqsimot xossasi

$$a(b + c) = ab + ac$$

$3(2x - 5) = 6x - 15$

Manfiy ishora qavsdagi **har bir** hadga tatbiq etiladi:

$-(3x - 5) \neq -3x - 5$ $\qquad -(3x - 5) = -3x + 5$

QOIDA · Ko'p hadli ifodani soddalashtirish

$4(x + 3) - 2(x - 1)$ ← 1. qavslarni oching

↓

$4x + 12 - 2x + 2$ ← 2. o'xshash hadlarni qo'shing

↓

$= 2x + 14$

Har bir qavs taqsimot xossasi bilan alohida ochiladi (manfiy ishoraga e'tibor bering), keyin o'xshash hadlar ixchamlanadi.

VOCABULARY – LUG'AT

Belgi	English	O'zbekcha
x	variable	*o'zgaruvchi*
x, y	variables	*o'zgaruvchilar*
–	term	*had*
–	terms	*hadlar*
$5x$	coefficient	*koeffitsient*
–	coefficients	*koeffitsientlar*
7	constant term	*erkin had*
c	constant	*doimiy*
–	like terms	*o'xshash hadlar*
–	combine like terms	*o'xshash hadlarni ixchamlash*
$a(b+c) = ab+ac$	distributive property	*taqsimot xossasi*
–	expression	*ifoda*
–	expressions	*ifodalar*
–	simplify	*soddalashtirish*
–	simplifies	*soddalashtiradi*
–	simplifying	*soddalashtirish*
=	equivalent	*teng kuchli (ekvivalent)*
( )	parentheses	*qavs*
–	negative	*manfiy*
+	positive	*musbat*
–	minus	*minus*
+	plus	*plyus*
–	value	*qiymat*
–	values	*qiymatlar*
=	equal	*teng*
–	error	*xato*
–	errors	*xatolar*
–	following	*quyidagi*
–	given	*berilgan*
–	represents	*ifodalaydi*

1-to'plam. *O'xshash hadlarni qo'shing (musbat koeffitsientlar).*

1. $3x + 5x =$

2. $7y + 2y + y =$

3. $4a + 3b + 2a =$

4. $6x + 3 + 2x =$

5. $5m + 4n + 3m + n =$

6. $2x + 3x + 4x =$

7. $8p + 5q + 2p + 3q =$

8. $x + x + x + x =$

2-to'plam. *O'xshash hadlarni qo'shing – ayirish ishtirok etadi.*

1. $7x - 3x =$

2. $5a - 8a =$

3. $4x - 7 + 2x =$

4. $-3y + 5y - 2y =$

5. $6m - 4n - 2m + 7n =$

6. $-5x + 3x - 8 =$

7. $2a - 5a + 6a =$

8. $9x - 4x - x =$

3-to'plam. *Taqsimot qonunini qo'llang (musbat ko'paytuvchi).*

1. $3(x+4) =$
2. $5(2x-3) =$
3. $2(3a+4b) =$
4. $4(x-1) =$
5. $6(2+3x) =$
6. $7(x+y) =$
7. $3(4x-2) =$
8. $2(5x+3) =$

4-to'plam. *Taqsimot qonuni – manfiy ko'paytuvchini har bir hadga tarqating.*

1. $-2(x+3) =$
2. $-(3x-5) =$
3. $-4(2x-1) =$
4. $-(x+7) =$
5. $-3(4-x) =$
6. $-(2x-3y) =$
7. $-5(x+2) =$
8. $-(-x+4) =$

5-to'plam. *Aralash – avval taqsimlang, so'ng o'xshash hadlarni qo'shing.*

1. $3(x+2)+4x=$

2. $5(2x-1)-3x=$

3. $2(x+3)-(x-1)=$

4. $4(x-2)+3(x+1)=$

5. $-(2x+3)+5x=$

6. $3(x-1)-2(x+2)=$

7. $2(3x+1)-(4x-5)=$

8. $5-2(x-3)=$

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri soddalashtirilgan ko'rinishni bering.*

1. $-(3x-5)$ ifodasini soddalashtirishda ko'rsatilgan yechimda xato bor:

   "$-3x-5$"

   Xato va to'g'ri javob: .....

2. $4x+3x^2$ ifodasini soddalashtirishda ko'rsatilgan yechimda xato bor:

   "$7x^2$"

   Xato va to'g'ri javob: .....

3. $-2(x-4)$ ifodasini soddalashtirishda ko'rsatilgan yechimda xato bor:

   "$-2x-8$"

   Xato va to'g'ri javob: .....

4. $5x-(2x-3)$ ifodasini soddalashtirishda ko'rsatilgan yechimda xato bor:

   "$5x-2x-3=3x-3$"

   Xato va to'g'ri javob: .....

5. $3(2x+1)-4$ ifodasini soddalashtirishda ko'rsatilgan yechimda xato bor:

   "$6x+1-4=6x-3$"

   Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – ichma-ich qavslar va ikki o'zgaruvchili ifodalar.*

1 $2\,[3(x+1)-2x]+5=$

2 $-3(x-2)+2(x+4)-5=$

3 $4x-2\,[x-(3x+1)]=$

4 $5(a-2b)-3(2a-b)=$

5 $-\,[2(x+3)-(x-1)]+4x=$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	$8x$	2	$10y$
	3	$6a + 3b$	4	$8x + 3$
	5	$8m + 5n$	6	$9x$
	7	$10p + 8q$	8	$4x$
2-to'plam	1	$4x$	2	$-3a$
	3	$6x - 7$	4	0
	5	$4m + 3n$	6	$-2x - 8$
	7	$3a$	8	$4x$
3-to'plam	1	$3x + 12$	2	$10x - 15$
	3	$6a + 8b$	4	$4x - 4$
	5	$18x + 12$	6	$7x + 7y$
	7	$12x - 6$	8	$10x + 6$
4-to'plam	1	$-2x - 6$	2	$5 - 3x$
	3	$4 - 8x$	4	$-x - 7$
	5	$3x - 12$	6	$-2x + 3y$
	7	$-5x - 10$	8	$x - 4$
5-to'plam	1	$7x + 6$	2	$7x - 5$
	3	$x + 7$	4	$7x - 5$
	5	$3x - 3$	6	$x - 7$
	7	$2x + 7$	8	$11 - 2x$
7-to'plam	1	$2x + 11$	2	$9 - x$
	3	$8x + 2$	4	$-a - 7b$
	5	$3x - 7$	–	–
6-to'plam	1. Xato – manfiy ishora faqat $3x$ ga tarqatilgan, -5 ga emas. To'g'ri: $-(3x - 5) = 5 - 3x$.			
	2. Xato – x va x^2 o'xshash had emas (daraja boshqa), qo'shib bo'lmaydi. To'g'ri: $3x^2 + 4x$ (allaqachon sodda).			
	3. Xato – $-2 \times (-4) = +8$ bo'lishi kerak edi, ishora tushib qolgan. To'g'ri: $-2(x - 4) = 8 - 2x$.			
	4. Xato – qavs oldidagi manfiy ishora faqat $2x$ ga tarqatilgan, -3 ga emas. To'g'ri: $5x - (2x - 3) = 3x + 3$.			
	5. Xato – 3×1 hisoblanmagan (ikkinchi hadga tarqatish o'tkazib yuborilgan). To'g'ri: $3(2x + 1) - 4 = 6x - 1$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram

FORMULA SHEET

QOIDA · Bir qadamli

$$x + 7 = 12$$
$$\downarrow \quad \leftarrow -7 \text{ ikkala tomonga}$$
$$x = 5$$

QOIDA · Ikki qadamli

$$3x - 5 = 10$$
$$\downarrow \quad \leftarrow +5$$
$$3x = 15$$
$$\downarrow \quad \leftarrow \div 3$$
$$x = 5$$

QOIDA · Ko’p qadamli: qavs ochish

Avval qavsni oching va o’xshash hadlarni ixchamlashtiring – shundan keyin oddiy ikki qadamli tenglama qoladi:

$$2(x - 4) + 5 = 7 \quad \leftarrow \text{1. qavs och, hadlarni qo’sh}$$
$$\downarrow$$
$$2x - 3 = 7 \quad \leftarrow \text{2. } +3 \text{ ikkala tomonga}$$
$$\downarrow$$
$$2x = 10 \quad \leftarrow \text{3. } \div 2 \text{ ikkala tomonga}$$
$$\downarrow$$
$$x = 5$$

QOIDA · Ikkala tomonda noma’lum

Noma’lumli hadlarni bir tomonga, sonli hadlarni ikkinchi tomonga yig’ing:

$$5x - 2 = 2x + 7 \quad \leftarrow \text{1. } -2x \text{ ikkala tomondan}$$
$$\downarrow$$
$$3x - 2 = 7 \quad \leftarrow \text{2. } +2 \text{ ikkala tomonga}$$
$$\downarrow$$
$$3x = 9 \quad \leftarrow \text{3. } \div 3 \text{ ikkala tomonga}$$
$$\downarrow$$
$$x = 3$$

VOCABULARY – LUG'AT

Belgi	English	O'zbekcha
=	equation	*tenglama*
–	equations	*tenglamalar*
–	solution	*yechim*
–	solutions	*yechimlar*
–	solve	*yechmoq*
–	solving	*yechish*
=	equal	*teng*
–	value	*qiymat*
–	values	*qiymatlar*
x	variable	*o'zgaruvchi*
–	variables	*o'zgaruvchilar*
–	coefficient	*koeffitsient*
–	coefficients	*koeffitsientlar*
–	constant	*doimiy*
–	constants	*o'zgarmaslar*
+	positive	*musbat*
–	negative	*manfiy*
–	error	*xato*
–	errors	*xatolar*
–	term	*had*
–	terms	*hadlar*
( )	parentheses	*qavs*
–	distributive property	*distributiv xossa*
–	given	*berilgan*
–	following	*quyidagi*
–	therefore	*demak*
–	true	*to'g'ri*
–	false	*noto'g'ri*
$\frac{a}{b}$	fraction	*kasr*
–	check (the answer)	*tekshirish*

1-to'plam. *Bir qadamli tenglamalar – asosiy amalni teskarisi bilan bajaring.*

1. $x + 7 = 12$
2. $x - 9 = 4$
3. $5x = 35$
4. $\frac{x}{4} = 6$
5. $x - (-3) = 10$
6. $-3x = 21$
7. $x + 15 = 8$
8. $\frac{x}{-5} = -6$

2-to'plam. *Ikki qadamli tenglamalar – qo'shish/ayirishdan so'ng ko'paytirish/bo'lish.*

1. $2x + 3 = 11$
2. $3x - 5 = 10$
3. $\frac{x}{2} + 4 = 9$
4. $5 - 2x = 1$
5. $4x + 7 = -5$
6. $-3x - 2 = 10$
7. $\frac{x}{3} - 1 = -4$
8. $7 - \frac{x}{2} = 3$

3-to'plam. *Qavsni ochish va o'xshash hadlarni birlashtirish.*

1. $3(x+2)=15$
2. $2(x-4)+5=7$
3. $4x+2x-3=15$
4. $-2(x+3)=10$
5. $3(2x-1)-2x=13$
6. $6-3(x-2)=0$
7. $2(3x+1)-(x-5)=22$
8. $-4(x-1)+2x=-2$

4-to'plam. *O'zgaruvchi ikkala tomonda – avval bir tomonga yig'ing.*

1. $3x+5=x+13$
2. $5x-2=2x+7$
3. $4x+1=2x-9$
4. $7-x=2x+1$
5. $2(x+3)=x+10$
6. $3x-4=5(x-2)$
7. $x+8=3x-2$
8. $6-2x=4x-18$

5-to'plam. *Aralash, SAT uslubida – ba'zilari x emas, hosila ifodani so'raydi. Diqqat bilan o'qing.*

1. $2x + 5 = 17$. $x + 3$ ning qiymati nechaga teng?

2. $3(x - 2) = 2x + 1$. $2x$ ning qiymati nechaga teng?

3. $\frac{x}{2} + 3 = 9$. $3x$ ning qiymati nechaga teng?

4. $4x - 7 = x + 8$. $x - 1$ ning qiymati nechaga teng?

5. $2(2x + 1) - 3 = 15$. x ning qiymati nechaga teng?

6. $5x + 2 = 3x + 12$. $5x$ ning qiymati nechaga teng?

7. $6 - x = 2x - 6$. $x + 4$ ning qiymati nechaga teng?

8. $3(x - 1) = x + 5$. $2x - 1$ ning qiymati nechaga teng?

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1. $x - 5 = 12$ ifodasida ko'rsatilgan yechimda xato bor:

   "1-qadam: ikkala tomonga 5 qo'shiladi: $x = 12$."

   Xato va to'g'ri javob: .....

2. $2x - 5 = 9$ ifodasida ko'rsatilgan yechimda xato bor:

   "1-qadam: $2x = 9 - 5 = 4$."

   Xato va to'g'ri javob: .....

3. $-2(x - 3) = 10$ ifodasida ko'rsatilgan yechimda xato bor:

   "1-qadam: $-2x - 6 = 10$."

   Xato va to'g'ri javob: .....

4. $3x + 5 = 20$ ifodasida ko'rsatilgan yechimda xato bor:

   "1-qadam: $8x = 20$, shuning uchun $x = 2.5$."

   Xato va to'g'ri javob: .....

5. $2x+1 = 11$ (savolda $3x$ so'ralgan) ifodasida ko'rsatilgan yechimda xato bor:

   "$x = 5$, javob: 5."

   Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – kasr va qavslar birgalikda, har biriga taxminan 90 soniya.*

1. $\frac{x+3}{2} = 7$

2. $\frac{3(x-1)}{2} = 6$

3. $2(x+1) - 3(x-2) = 10$

4. $\frac{2x-1}{3} + 1 = 4$

5. $4(x-2) = 2(x+3) - 6$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	5	2	13
	3	7	4	24
	5	7	6	-7
	7	-7	8	30
2-to'plam	1	4	2	5
	3	10	4	2
	5	-3	6	-4
	7	-9	8	8
3-to'plam	1	3	2	5
	3	3	4	-8
	5	4	6	4
	7	3	8	3
4-to'plam	1	4	2	3
	3	-5	4	2
	5	4	6	3
	7	5	8	4
5-to'plam	1	9	2	14
	3	36	4	4
	5	4	6	25
	7	8	8	7
7-to'plam	1	11	2	5
	3	-2	4	5
	5	4	–	–
6-to'plam	1. 5 faqat o'ng tomonga qo'shilgan – ikkala tomonga qo'shish kerak. To'g'ri: $x = 12 + 5 = 17$.			
	2. -5 ishorasi almashtirilmagan. To'g'ri: $2x = 9 + 5 = 14$, $x = 7$.			
	3. -2 faqat x ga taqsimlangan, -3 ga emas. To'g'ri: $-2x + 6 = 10 \Rightarrow x = -2$.			
	4. $3x$ va 5 o'xshash had emas. To'g'ri: $3x = 15 \Rightarrow x = 5$.			
	5. $x = 5$ to'g'ri, lekin savol $3x$ ni so'ragan. To'g'ri: $3x = 15$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram

FORMULA SHEET

QOIDA · Bir xil asosli daraja qoidalari

Qoida	Misol
$a^m \times a^n = a^{m+n}$	$2^3 \times 2^4 = 2^{3+4} = 2^7$
$(a^m)^n = a^{mn}$	$(2^7)^2 = 2^{7\times 2} = 2^{14}$
$\frac{a^m}{a^n} = a^{m-n}$	$\frac{2^{14}}{2^5} = 2^{14-5} = 2^9$

QOIDA · Nol va manfiy daraja

Nolmas a uchun: $a^0 = 1$. Manfiy ko'rsatkich – teskari sonni bildiradi, asosni manfiy qilmaydi:

$$5^0 + 2^{-3} \times 4^2$$

← 1. $a^0=1$, $a^{-n}=\frac{1}{a^n}$

$$\downarrow$$

$$1 + \frac{1}{8} \times 16$$

← 2. ko'paytiring

$$\downarrow$$

$$1 + 2 \qquad = 3$$

QOIDA · Kvadrat ildizni soddalashtirish

Eng katta to'liq kvadrat ko'paytuvchini toping, so'ng ajrating: $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$.

$$\sqrt{72}$$

← 1. $72=36\times 2$, 36 – to'liq kvadrat

$$\downarrow$$

$$\sqrt{36 \times 2} = \sqrt{36} \times \sqrt{2}$$

← 2. ajrating

$$\downarrow$$

$$= 6\sqrt{2}$$

Eslatma: $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$ *– masalan* $\sqrt{16+9} = 5$, *biroq* $\sqrt{16} + \sqrt{9} = 7$.

VOCABULARY – LUG'AT

Belgi	English	O'zbekcha
x^2	exponent / power	*daraja*
a (in a^n)	base	*asos*
n (in a^n)	exponent (index)	*ko'rsatkich*
a^0	zero exponent	*nol darajasi*
a^{-n}	negative exponent	*manfiy daraja*
$\frac{1}{a}$	reciprocal	*teskari son*
$\sqrt{a}$	square root	*kvadrat ildiz*
$36 = 6^2$	perfect square	*to'liq kvadrat*
–	value	*qiymat*
=	equal	*teng*
≡	equivalent	*teng kuchli*
+	positive	*musbat*
–	negative	*manfiy*
–	real	*haqiqiy*
$\mathbb{Z}$	integer	*butun son*
$\frac{a}{b}$	fraction	*kasr*
–	solve	*yechmoq*
–	solution	*yechim*
–	equation	*tenglama*
–	expression	*ifoda*
–	simplifying	*soddalashtirish*
–	error	*xato*
–	factor	*ko'paytuvchi*
x	variable	*o'zgaruvchi*
x^2	square	*kvadrat*
$\sqrt{}$	root	*ildiz*

1-to'plam. *Daraja qoidalari – nol/manfiy ko'rsatkichsiz sof mashq.*

1. $2^2 \times 2^3 =$

2. $\frac{3^4}{3^2} =$

3. $(2^2)^3 =$

4. $5^3 \times 5^2 =$

5. $\frac{4^5}{4^3} =$

6. $(3^2)^2 =$

7. $2^3 \times 2^3 =$

8. $(2^3)^3 =$

2-to'plam. *Nol va manfiy ko'rsatkichlar.*

1. $5^0 =$

2. $2^{-3} =$

3. $4^{-2} =$

4. $2^0 \times 2^{-2} =$

5. $3^{-2} \times 3^4 =$

6. $5^2 \times 5^{-2} =$

7. $10^{-1} \times 100 =$

8. $2^{-1} + 2^{-2} =$

3-to'plam. *Aralash – haqiqiy test zichligi.*

1. $\dfrac{(2^2 \times 2^3)^2}{2^4} =$

2. $(3 \times 3^2)^2 =$

3. $\dfrac{5^3}{5^2 \times 5^{-1}} =$

4. $(2^{-1})^{-3} =$

5. $(4^2)^0 =$

6. $\dfrac{2^5}{2^{-2}} =$

7. $\dfrac{(3^2)^2}{3^3} =$

8. $10^3 \times 10^{-1} \div 10 =$

4-to'plam. *Kvadrat ildizlarni soddalashtirish.*

1. $\sqrt{16} =$

2. $\sqrt{49} =$

3. $\sqrt{18} =$

4. $\sqrt{50} =$

5. $\sqrt{72} =$

6. $\sqrt{20} =$

7. $\sqrt{45} =$

8. $\sqrt{8} =$

5-to'plam. *Ildizlarni qo'shish va ayirish.*

1. $\sqrt{8} + \sqrt{2} =$

2. $\sqrt{18} - \sqrt{2} =$

3. $\sqrt{12} + \sqrt{27} =$

4. $\sqrt{50} - \sqrt{8} =$

5. $2\sqrt{3} \times \sqrt{3} =$

6. $\sqrt{2} \times \sqrt{8} =$

7. $3\sqrt{5} + 2\sqrt{5} =$

8. $\sqrt{45} - \sqrt{20} =$

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1. $2^3 \times 2^3$ ifodasida ko'rsatilgan yechimda xato bor:
   "$2^3 \times 2^3 = 2^9 = 512$."
   Xato va to'g'ri javob: .....

2. $(2^2)^3$ ifodasida ko'rsatilgan yechimda xato bor:
   "$(2^2)^3 = 2^{2+3} = 2^5 = 32$."
   Xato va to'g'ri javob: .....

3. 2^{-3} ifodasida ko'rsatilgan yechimda xato bor:
   "$2^{-3} = -8$."
   Xato va to'g'ri javob: .....

4. $\sqrt{16 + 9}$ ifodasida ko'rsatilgan yechimda xato bor:
   "$\sqrt{16} + \sqrt{9} = 4 + 3 = 7$."
   Xato va to'g'ri javob: .....

5. $\sqrt{32}$ ifodasida ko'rsatilgan yechimda xato bor:
   "$32 = 16 \times 2$ *deb to'g'ri ajratilgan, lekin javobda* $16\sqrt{2}$ *deb yozilgan.*"
   Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – har biriga taxminan 90 soniya.*

1. $\dfrac{(2^3)^2}{2^4} + \sqrt{16} =$

2. $\sqrt{3^2 \times 2^2} =$

3. $2^{-2} \times 4^2 + \sqrt{9} =$

4. $(\sqrt{2})^4 =$

5. $1 + \sqrt{5^2 - 3^2} =$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	32	2	9
	3	64	4	3125
	5	16	6	81
	7	64	8	512
2-to'plam	1	1	2	$\frac{1}{8}$
	3	$\frac{1}{16}$	4	$\frac{1}{4}$
	5	9	6	1
	7	10	8	$\frac{3}{4}$
3-to'plam	1	64	2	729
	3	25	4	8
	5	1	6	128
	7	3	8	10
4-to'plam	1	4	2	7
	3	$3\sqrt{2}$	4	$5\sqrt{2}$
	5	$6\sqrt{2}$	6	$2\sqrt{5}$
	7	$3\sqrt{5}$	8	$2\sqrt{2}$
5-to'plam	1	$3\sqrt{2}$	2	$2\sqrt{2}$
	3	$5\sqrt{3}$	4	$3\sqrt{2}$
	5	6	6	4
	7	$5\sqrt{5}$	8	$\sqrt{5}$
7-to'plam	1	8	2	6
	3	7	4	4
	5	5	–	–
6-to'plam	1. Ko'rsatkichlarni ko'paytirgan, qo'shish kerak edi (ko'paytma qoidasi). To'g'ri: $2^{3+3} = 2^6 = 64$.			
	2. Ko'rsatkichlarni qo'shgan, ko'paytirish kerak edi (daraja darajasi). To'g'ri: $2^{2\times3} = 2^6 = 64$.			
	3. Manfiy ko'rsatkichni asosni manfiy qiladi deb tushungan. To'g'ri: $2^{-3} = \frac{1}{8}$.			
	4. Yig'indining ildizini ildizlar yig'indisiga bo'lgan – bu hech qachon to'g'ri emas. To'g'ri: $\sqrt{25} = 5$.			
	5. 16 ni to'g'ri ajratgan, lekin uning ildizini olmagan. To'g'ri: $\sqrt{16 \times 2} = 4\sqrt{2}$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

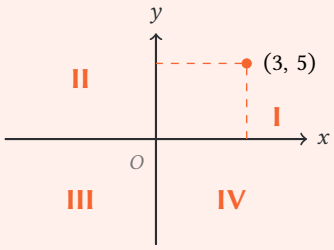
YouTube

QR kod

Telegram

FORMULA SHEET

QOIDA · Koordinatalar tekisligi



Chorak	x	y
I	+	+
II	−	+
III	−	−
IV	+	−

QOIDA · $y = kx + b$

$k = -3$ **(qiyalik/slope)**

$$y = -3x + 5$$

$b = 5$ **(Oy-kesma)**

SAT'da k o'rniga m harfi va "slope" so'zi ishlatiladi: $y = mx + b$. $k = m$ – faqat nom boshqa.

QOIDA · $f(x)$ yozuvi

$f(x)$ – bu shunchaki y ning boshqa nomi. x o'rniga son qo'yib, ifodani hisoblang:

$$f(x) = 2x + 3$$
$$f(4) = 2(4) + 3$$
$$= 8 + 3$$
$$f(4) = 11$$

$f(x) = x^2 - 2x$

← manfiy son qavs ichida!

$f(-3) = \underline{(-3)}^2 - 2\underline{(-3)}$

$= 9 + 6 = 15$

VOCABULARY – LUG'AT

Belgi	English	O'zbekcha
(x, y)	**point**	*nuqta*
–	**points**	*nuqtalar*
–	**coordinate**	*koordinata*
(x, y)	**coordinates**	*koordinatalar*
–	**plane**	*tekislik*
–	**axis**	*o'q*
$(0, 0)$	**origin**	*koordinata boshi*
–	**quadrant**	*chorak (kvadrant)*
–	**linear**	*chiziqli*
$f(x)$	**function**	*funksiya*
–	**functions**	*funksiyalar*
–	**value**	*qiymat*
–	**values**	*qiymatlar*
–	**negative**	*manfiy*
+	**positive**	*musbat*
–	**equation**	*tenglama*
–	**equations**	*tenglamalar*
m	**slope**	*qiyalik*
b	**intercept**	*kesma*
–	**graph**	*grafik*
–	**line**	*chiziq*
–	**given**	*berilgan*
–	**represents**	*ifodalaydi*
–	**substituting**	*o'rniga qo'yish*
–	**constant**	*doimiy*
a	**coefficient**	*koeffitsient*
–	**horizontal**	*gorizontal*
–	**vertical**	*vertikal*
x^2	**squared**	*kvadratga ko'tarilgan*

1-to'plam. *Nuqta qaysi chorakka tegishli ekanini aniqlang.*

1. $(3,\ 5)$ → chorak:

2. $(-2,\ 4)$ → chorak:

3. $(-6,\ -1)$ → chorak:

4. $(7,\ -3)$ → chorak:

5. $(-4,\ -8)$ → chorak:

6. $(5,\ -5)$ → chorak:

7. $(-9,\ 2)$ → chorak:

8. $(1,\ 1)$ → chorak:

2-to'plam. *Tenglamadan k (burchak koeffitsienti) va b ni ajrating.*

1. $y = 3x + 7$ → $k=$ $b=$

2. $y = -2x + 1$ → $k=$ $b=$

3. $y = x - 5$ → $k=$ $b=$

4. $y = -x + 9$ → $k=$ $b=$

5. $y = 4x$ → $k=$ $b=$

6. $y = -6$ → $k=$ $b=$

7. $y = 0.5x - 3$ → $k=$ $b=$

8. $y = \frac{1}{3}x - 2$ → $k=$ $b=$

3-to'plam. *$f(a)$ ni musbat qiymatlar uchun hisoblang.*

1. $f(x) = 2x + 3.\ f(4) =$

2. $f(x) = 5x - 1.\ f(2) =$

3. $f(x) = -3x + 4.\ f(1) =$

4. $f(x) = x + 10.\ f(0) =$

5. $f(x) = 4x.\ f(3) =$

6. $f(x) = -x + 6.\ f(5) =$

7. $f(x) = 2x - 7.\ f(10) =$

8. $f(x) = -5x + 2.\ f(3) =$

4-to'plam. *$f(a)$ ni manfiy qiymatlar uchun hisoblang – manfiy son har doim qavsda.*

1. $f(x) = 3x + 2.\ f(-2) =$

2. $f(x) = -4x + 1.\ f(-3) =$

3. $f(x) = x + 8.\ f(-8) =$

4. $f(x) = x^2.\ f(-3) =$

5. $f(x) = x^2 + 1.\ f(-4) =$

6. $f(x) = 2x^2 - 3.\ f(-2) =$

7. $f(x) = -x^2.\ f(-5) =$

8. $f(x) = (x + 1)^2.\ f(-6) =$

5-to'plam. *Aralash – shaklni almashtiring, qiymatni hisoblang, tenglamani yeching.*

1 $y = 5x - 3$ ni $f(x)$ shaklida yozing va $f(2)$ ni toping.

2 $f(x) = -2x + 8$ ni $y = kx + b$ shaklida yozing, k, b ni toping.

3 $y = x + 1$ ni $f(x)$ shaklida yozing, so'ng $f(-4)$ ni hisoblang.

4 $f(x) = 4x - 9$. Agar $f(x) = 11$ bo'lsa, x ni toping.

5 $f(x) = -3x + 6$. Agar $f(x) = 0$ bo'lsa, x ni toping.

6 $y = -x - 5$ ni $f(x)$ shaklida yozing va $f(0)$ ni toping.

7 $f(x) = 2x + 1$ bo'lsa, $f(3) + f(-3)$ ni hisoblang.

8 $f(x) = x^2 - 4$. Agar $f(a) = 5$ bo'lsa, a ning musbat qiymatini toping.

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'g'ri javobni bering.*

1 $f(x) = 3x + 2$. $f(5)$ ni toping:
"$f(5) = f \cdot 5$, f *noma'lum, javob topib bo'lmaydi.*"
Xato va to'g'ri javob: .....

2 $f(x) = x^2$. $f(-4)$ ni toping:
"$f(-4) = -4^2 = -16$."
Xato va to'g'ri javob: .....

3 $(-3,\ 5)$ nuqtasi qaysi chorakda?
"$x = -3$ *ni musbat deb o'ylab, I chorak.*"
Xato va to'g'ri javob: .....

4 $f(x) = 2x - 5$. Agar $f(x) = 7$, x ni toping:
"$2x - 5 = 7 \Rightarrow 2x = 2 \Rightarrow x = 1$."
Xato va to'g'ri javob: .....

5 $f(x) = x^2 + 3$. $f(-2)$ ni toping:
"$f(-2) = -2^2 + 3 = -4 + 3 = -1$."
Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin – har biriga taxminan 90 soniya.*

1 $f(x) = 2x^2 - 3x + 1$. $f(-2) =$

2 $f(x) = -(x-3)^2$. $f(6) =$

3 $y = -4x + 11$ ni $f(x)$ shaklida yozing, so'ng $f(x) = 3$ bo'lganda x ni toping.

4 $f(x) = x^2 - 2x$. Agar $f(a) = 8$ bo'lsa, a ning manfiy qiymatini toping.

5 $f(x) = 3x + c$. Agar $f(2) = 10$ bo'lsa, c ni toping.

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	I	2	II
	3	III	4	IV
	5	III	6	IV
	7	II	8	I
2-to'plam	1	k=3, b=7	2	k=−2, b=1
	3	k=1, b=−5	4	k=−1, b=9
	5	k=4, b=0	6	k=0, b=−6
	7	k=0.5, b=−3	8	$k=\frac{1}{3}$, b=−2
3-to'plam	1	11	2	9
	3	1	4	10
	5	12	6	1
	7	13	8	−13
4-to'plam	1	−4	2	13
	3	0	4	9
	5	17	6	5
	7	−25	8	25
5-to'plam	1. $f(x) = 5x-3$; $f(2) = 7$ 2. $y = -2x+8$; $k = -2$, $b = 8$			
	3. $f(x) = x + 1$; $f(-4) = -3$ 4. $x = 5$			
	5. $x = 2$ 6. $f(x) = -x - 5$; $f(0) = -5$			
	7. $f(3) + f(-3) = 7 + (-5) = 2$			
	8. $a = 3$			
7-to'plam	1	15	2	−9
	3. $f(x) = -4x + 11$; $x = 2$ 4. $a = -2$ 5. $c = 4$			
6-to'plam	1. f funksiya nomi, songa ko'paytirilmaydi. To'g'ri: $f(5) = 3(5) + 2 = 17$.			
	2. Manfiy kiritmada qavs tushib qolgan. To'g'ri: $f(-4) = (-4)^2 = 16$.			
	3. x ning ishorasi noto'g'ri o'qilgan. To'g'ri: $x = -3<0$, $y=5>0 \Rightarrow$ II chorak.			
	4. $7 + 5 = 12$, 2 emas. To'g'ri: $2x = 12 \Rightarrow x = 6$.			
	5. Yana qavs tushib qolgan. To'g'ri: $f(-2) = (-2)^2+3 = 7$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram

FORMULA SHEET

QOIDA · 1–8-darslar – asosiy formulalar

#	Mavzu	Asosiy qoida
1	Amallar tartibi	Qavs → Daraja/Ildiz → ×÷ → +−
2	Kasrlar	$\frac{a}{b} \pm \frac{c}{b}$: umumiy maxraj; $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$
3	O'nli kasr / foiz	%→o'nli kasr: ÷100; o'nli kasr → %: ×100
4	Ishorali sonlar	Xil ishora: ayirib, kattaning ishorasi. (×, ÷): bir xil ishora → +, har xil → −
5	Ifodalar	$a(b + c) = ab + ac$; o'xshash hadlar: koeffitsientlar qo'shiladi
6	Tenglamalar	Ikkala tomonga bir xil teskari amalni qo'llang
7	Daraja / ildiz	$a^m \cdot a^n = a^{m+n}$; $\frac{a^m}{a^n} = a^{m-n}$; $a^0 = 1$
8	Koordinata / funksiya	$y = kx + b \equiv f(x) = kx + b$; qiyalik $= k$

VOCABULARY – LUG'AT

Belgi	English	O'zbekcha
=	**equation**	*tenglama*
=	**equations**	*tenglamalar*
–	**expression**	*ifoda*
–	**expressions**	*ifodalar*
x	**variable**	*o'zgaruvchi*
x, y	**variables**	*o'zgaruvchilar*
a	**constant**	*doimiy*
a, b	**constants**	*o'zgarmaslar*
–	**term**	*had*
–	**terms**	*hadlar*
$\frac{a}{b}$	**fraction**	*kasr*
0.5	**decimal**	*o'nlik*
a (in $\frac{a}{b}$)	**numerator**	*surat*
b (in $\frac{a}{b}$)	**denominator**	*maxraj*
%	**percent**	*foiz*
%	**percentage**	*foiz*
–	**negative**	*manfiy*
+	**positive**	*musbat*
x^2	**exponent**	*daraja*
x^n	**exponents**	*daraja*
$\sqrt{x}$	**root**	*ildiz*
$f(x)$	**function**	*funksiya*
$f(x), g(x)$	**functions**	*funksiyalar*
–	**solve**	*yechmoq*
–	**solving**	*yechish*
–	**solution**	*yechim*
–	**solutions**	*yechimlar*
–	**value**	*qiymat*
–	**order**	*tartib*
–	**error**	*xato*

1-to'plam. *Amallar tartibi va kasrlarni birlashtiring.*

1. $\frac{3}{4} + 2 \times \left(\frac{1}{2} - \frac{1}{8}\right) =$

2. $\left(\frac{2}{3} + \frac{1}{6}\right) \times 4 - 1 =$

3. $10 - 3 \times \left(\frac{2}{5} + \frac{1}{10}\right) =$

4. $\left(\frac{7}{8} - \frac{1}{4}\right) \div \frac{1}{2} =$

5. $2 \times \frac{3}{4} + \left(1 - \frac{1}{3}\right) \times 6 =$

6. $\left(\frac{1}{2} + \frac{1}{3} + \frac{1}{6}\right) \times 12 - 5 =$

2-to'plam. *Ishorali sonlar va darajalarni birlashtiring.*

1. $(-2)^3 - 3 \times (-4) =$

2. $-3^2 + (-3)^2 =$

3. $(-5)^2 \div (-5) + 2^3 =$

4. $-2 \times (-3)^2 + (-4) =$

5. $(-1)^5 \times (-2)^3 =$

6. $4 - (-2)^4 \div 2 =$

3-to'plam. *O'nli kasr/foiz va tenglamalarni birlashtiring. x ni toping.*

1. $0.25x + 10 = 35$

2. 40% of x is 32

3. $x - 15\%$ of $x = 68$

4. $0.5(x + 4) = 13$

5. 20% of $(x + 10) = 16$

6. $x + 0.1x = 44$

4-to'plam. *Algebraik ifodalar va tenglamalarni birlashtiring. x ni toping.*

1. $3(x - 2) + 5 = 2x + 7$

2. $2(x + 3) - (x - 1) = 10$

3. $4x - (2x + 5) = 7$

4. $5(x - 1) = 3(x + 3)$

5. $-2(x + 4) + 6 = x - 2$

6. $3(2x - 1) - 4 = 2(x + 5)$

5-to'plam. *Kasrli tenglamalarni yeching. x ni toping.*

1. $\frac{2}{3}x - 1 = 5$

2. $\frac{x}{4} + 3 = 7$

3. $\frac{3}{4}x + \frac{1}{2} = 2$

4. $\frac{x+2}{3} = 5$

5. $\frac{2x-1}{5} = 3$

6. $\frac{x}{2} - \frac{x}{3} = 2$

6-to'plam. *$f(x)$ funksiyasi, ishorali sonlar va darajalar.*

1. $f(x) = -2x^2 + 3$. $f(-2) =$

2. $f(x) = x^3 - 4x$. $f(-1) =$

3. $f(x) = (x-2)^2$. $f(-3) =$

4. $f(x) = 2x^2 - 5x + 1$. $f(-1) =$

5. $f(x) = -x^2 + 2x$. $f(3) =$? $f(-3) =$

6. $f(x) = kx + b$. $f(2) = 7, f(0) = 3$. $k =$? $b =$

7-to'plam. *Xatoni toping – ikki mavzu aralashgan, bitta xato qadam bor.*

1 $\frac{1}{2}x + 3 = 7$ ifodasida ko'rsatilgan yechimda xato bor:

"$\frac{1}{2}x = 7 + 3 = 10$, *shuning uchun* $x = 20$."

Xato va to'g'ri javob: .....

2 $3(x - 2) = 9$ ifodasida ko'rsatilgan yechimda xato bor:

"$3x - 2 = 9$, *shuning uchun* $3x = 11$, $x = \frac{11}{3}$."

Xato va to'g'ri javob: .....

3 $f(x) = x^2 - 2x$, $f(-3)$ ifodasida ko'rsatilgan yechimda xato bor:

"$f(-3) = (-3)^2 - 2(-3) = 9 - 6 = 3$."

Xato va to'g'ri javob: .....

4 $0.4x = 24$ ifodasida ko'rsatilgan yechimda xato bor:

"$x = 24 \times 0.4 = 9.6$."

Xato va to'g'ri javob: .....

5 $2(x + 1) - x = 9$ ifodasida ko'rsatilgan yechimda xato bor:

"$2x + 1 - x = 9$, $x + 1 = 9$, $x = 8$."

Xato va to'g'ri javob: .....

8-to'plam. *Bonus / eng qiyin – bitta masalada 3+ ko'nikma.*

1 $\frac{1}{2}(x - 4) + 0.25x = 5$

2 $f(x) = -2(x - 1)^2 + 3$. $f(-1) =$

3 30% of $(2x + 10) = x + 2$

4 $\frac{3}{4}x - (-2)^3 = 11$

5 $f(x) = kx + b$. $f(-2) = 1$, $f(2) = 9$. $k =$? $b =$? $f(0) =$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	$\frac{3}{2}$	2	$\frac{7}{3}$
	3	$\frac{17}{2}$	4	$\frac{5}{4}$
	5	$\frac{11}{2}$	6	7
2-to'plam	1	4	2	0
	3	3	4	-22
	5	8	6	-4
3-to'plam	1	$x=100$	2	$x=80$
	3	$x=80$	4	$x=22$
	5	$x=70$	6	$x=40$
4-to'plam	1	$x=8$	2	$x=3$
	3	$x=6$	4	$x=7$
	5	$x=0$	6	$x=\frac{17}{4}$
5-to'plam	1	$x=9$	2	$x=16$
	3	$x=2$	4	$x=13$
	5	$x=8$	6	$x=12$
6-to'plam	1	-5	2	3
	3	25	4	8
	5	$f(3)=-3$, $f(-3)=-15$		
	6	$k=2$, $b=3$		
7-to'plam	1. Teskari amal xato – 3 qo'shish o'rniga ayirish kerak edi. To'g'ri: $\frac{1}{2}x = 4$, $x = 8$.			
	2. 3 faqat x ga taqsimlangan, -2 ga emas. To'g'ri: $3x - 6 = 9$, $x = 5$.			
	3. $-2 \times (-3) = -6$ emas, $+6$. To'g'ri: $9 + 6 = 15$.			
	4. Ko'paytirish o'rniga bo'lish kerak edi. To'g'ri: $x = 24 \div 0.4 = 60$.			
	5. 2 faqat x ga taqsimlangan, $+1$ ga emas. To'g'ri: $2x + 2 - x = 9$, $x = 7$.			
8-to'plam	1	$x=\frac{28}{3}$		
	2	-5	3	$x=\frac{5}{2}$
	4	$x=4$	5	$k=2, b=5$

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
about	*taxminan*
absolute	*mutlaq*
according	*ga ko’ra*
account	*hisob*
activity	*faoliyat*
actual	*haqiqiy*
acute	*o’tkir*
added	*qo’shilgan*
adding	*qo’shish*
additional	*qo’shimcha*
additionally	*bundan tashqari*
adjacent	*qo’shni*
adult	*katta yoshli*
adults	*kattalar*
advanced	*ilg’or*
against	*qarshi*
age	*yosh*
air	*havo*
algebra	*algebra*
already	*allaqachon*
alternate	*almashinuvchi*
altitude	*balandlik*
amount	*miqdor*
analysis	*tahlil*
angle	*burchak*

English	O’zbekcha
angles	*burchaklar*
animal	*hayvon*
animals	*hayvonlar*
annual	*yillik*
answer	*javob*
appears	*ko’rinadi*
applied	*qo’llanildi*
applying	*qo’llash*
approach	*yondashuv*
appropriate	*mos*
approximate	*taxminiy*
approximately	*taxminan*
april	*aprel*
arc	*yoy*
area	*yuza*
areas	*yuzalar*
article	*maqola*
ascending	*o’suvchi (tartib)*
asked	*so’ralgan*
assigned	*tayinlangan*
associated	*bog’liq*
assuming	*faraz qilib*
attended	*qatnashgan*
attendee	*ishtirokchi*
attendees	*ishtirokchilar*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
auger	*parma*
august	*avgust*
average	*o’rtacha*
away	*uzoqlikda*
axis	*o’q*
azaleas	*azaliya*
bacteria	*bakteriyalar*
bag	*xalta*
bags	*sumkalar*
balance	*balans*
ball	*to’p*
bank	*bank*
bar	*ustun*
base	*asos*
based	*asoslangan*
batch	*partiya*
beach	*plyaj*
beads	*munchoqlar*
beetle	*qo’ng’iz*
began	*boshlamoq*
best	*eng yaxshi*
bicycle	*velosiped*
biked	*velosipedda yurdi*
biking	*velosiped haydash*
bill	*hisob (to’lov)*

English	O’zbekcha
bin	*quti*
birch	*qayin*
bird	*qush*
black	*qora*
blackberries	*yejevika*
blood	*qon*
blue	*ko’k*
board	*taxta*
boards	*taxtalar*
boat	*qayiq*
body	*tana*
boiling	*qaynash*
book	*kitob*
books	*kitoblar*
born	*tug’ilgan*
bottles	*shishalar*
bottom	*tag*
bought	*sotib oldi*
boundary	*chegara*
bowl	*kosa*
box	*quti*
boxes	*qutilar*
boxwoods	*buksus*
brakes	*tormozlar*
budget	*byudjet*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
buildings	*binolar*
bursts	*yoriladi*
bus	*avtobus*
bushels	*bushel*
business	*biznes*
button	*tugma*
buttons	*tugmalar*
buy	*sotib olmoq*
buys	*sotib oladi*
calcium	*kaltsiy*
calculated	*hisoblangan*
calculating	*hisoblash*
calculation	*hisoblash*
calories	*kaloriyalar*
camping	*kemping*
candle	*sham*
candles	*shamlar*
cannot	*mumkin emas*
capacity	*sig’im*
car	*avtomobil*
card	*karta*
cards	*kartalar*
cargo	*yuk*
cars	*mashina*
cell	*katak*

English	O’zbekcha
cells	*katakcha*
celsius	*selsiy*
center	*markaz*
centimeter	*santimetr*
centimeters	*santimetr*
cereal	*donli taom*
certain	*ma’lum (bir)*
change	*o’zgarish*
charge	*to’lov*
charged	*to’lov olingan*
charges	*to’lovlar*
cheese	*pishloq*
child	*bola*
children	*bolalar*
chips	*chipslar*
chocolate	*shokolad*
chose	*tanladi*
chosen	*tanlangan*
circle	*doira*
circular	*doiraviy*
circumference	*aylana uzunligi*
city	*shahar*
class	*sinf*
classes	*toifalar*
clients	*mijozlar*

VOCABULARY – COMPLETE GLOSSARY

English	O'zbekcha
closest	*eng yaqin*
club	*to'garak*
cluster	*klaster*
coefficient	*koeffitsient*
coefficients	*koeffitsientlar*
coin	*tanga*
coins	*tangalar*
collect	*to'plamoq*
collected	*to'plandi*
collection	*to'plam*
colony	*koloniya*
color	*rang*
colored	*rangli*
combinations	*kombinatsiyalar*
combined	*qo'shilgan*
combining	*birlashtirish*
comic	*komiks*
comments	*sharhlar*
common	*umumiy*
company	*kompaniya*
compare	*solishtirmoq*
compares	*solishtiradi*
complete	*to'ldirmoq*
completed	*bajarilgan*
computational	*hisoblash*

English	O'zbekcha
computer	*kompyuter*
conceptual	*tushunchaviy*
conclusion	*xulosa*
concrete	*beton*
condition	*shart*
conditional	*shartli*
cone	*konus*
congruent	*teng (kongruent)*
consecutive	*ketma-ket*
consisting	*...dan iborat*
consists	*iborat*
constant	*doimiy*
constants	*o'zgarmaslar*
constructed	*qurilgan*
contain	*o'z ichiga olmoq*
container	*idish*
containers	*idishlar*
containing	*o'z ichiga olgan*
contains	*o'z ichiga oladi*
context	*kontekst*
cookies	*pechenye*
coordinate	*koordinata*
coordinates	*koordinatalar*
corn	*makkajo'xori*
correctly	*to'g'ri*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
correspond	*mos kelmoq*
corresponding	*mos*
corresponds	*mos keladi*
cosine	*kosinus*
cost	*narx*
costs	*turadi (narxi ... bo’ladi)*
country	*mamlakat*
courses	*kurs*
crackers	*krekerlar*
crates	*yashiklar*
cream	*qaymoq*
create	*yaratmoq*
created	*yaratilgan*
crosses	*kesib o’tadi*
cube	*kub*
cubes	*kub*
cubic	*kubik*
cup	*stakan*
cups	*stakanlar*
customer	*mijoz*
customers	*mijozlar*
cutting	*kesish*
cylinder	*silindr*
daily	*kunlik*
dance	*raqs*

English	O’zbekcha
data	*ma’lumotlar*
day	*kun*
daylight	*kun yorug’ligi*
days	*kunlar*
dealer	*sotuvchi*
decay	*kamayish*
decimal	*o’nlik*
decrease	*kamayish*
decreased	*kamaymoq*
decreases	*kamayadi*
decreasing	*kamayuvchi*
defective	*nuqsonli*
define	*ta’riflamoq*
defined	*aniqlangan*
defines	*aniqlaydi*
defining	*aniqlovchi*
definition	*ta’rif*
degree	*gradus*
degrees	*gradus*
deliveries	*yetkazib berishlar*
denominator	*maxraj*
density	*zichlik*
department	*bo’lim*
deposit	*depozit*
deposits	*depozitlar*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
depth	*chuqurlik*
descending	*kamayish tartibida*
describes	*tasvirlamoq*
designer	*dizayner*
determine	*aniqlamoq*
determined	*aniqlangan*
determining	*aniqlash*
deviation	*chetlanish*
device	*qurilma*
diagonal	*diagonal*
diameter	*diametr*
difference	*ayirma*
different	*turli*
discount	*chegirma*
discounted	*chegirma qilingan*
discriminant	*diskriminant*
dispenser	*tarqatgich*
displays	*ko’rsatadi*
distance	*masofa*
distinct	*farqli*
distributing	*qavsni ochish*
distribution	*taqsimot*
distributions	*taqsimotlar*
distributive	*taqsimot*
diver	*sho’ng’uvchi*

English	O’zbekcha
divided	*bo’lingan*
dividing	*bo’lish*
doesn’t	*...emas (qilmaydi)*
dollars	*dollar*
dot	*nuqta*
double	*ikki baravar*
doubles	*ikki baravar ko’payadi*
dragonflies	*ninachilar*
drawing	*chizma*
driven	*haydalgan*
dropped	*tashlangan*
drove	*haydadi*
earn	*ishlab topmoq*
earned	*ishlab topdi*
earnings	*daromad*
earns	*ishlab topadi*
edge	*qirra*
egg	*tuxum*
eggs	*tuxumlar*
eighth	*sakkizinchi*
electric	*elektr*
elements	*elementlar*
elevation	*balandlik*
employee	*xodim*
employees	*xodimlar*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
end	*oxiri*
energy	*energiya*
enrolled	*ro’yxatdan o’tmoq*
enter	*kiritish*
entire	*butun*
equal	*teng*
equation	*tenglama*
equations	*tenglamalar*
equilateral	*teng tomonli*
equipment	*jihozlar*
equivalent	*teng kuchli*
error	*xato*
errors	*xatolar*
essay	*insho*
estimate	*baholamoq (taxmin qilish)*
estimated	*taxmin qilingan*
estimates	*baholaydi*
evaluating	*hisoblash*
even	*juft*
event	*hodisa*
exactly	*aynan*
example	*misol*
examples	*misollar*
exerted	*qo’llangan (kuch)*
expanding	*yoyish*

English	O’zbekcha
expected	*kutilgan*
expenses	*xarajatlar*
experiment	*tajriba*
exponent	*daraja*
exponential	*ko’rsatkichli*
exponents	*daraja*
express	*ifodalamoq*
expressed	*ifodalangan*
expresses	*ifodalaydi*
expression	*ifoda*
expressions	*ifodalar*
exterior	*tashqi*
fabric	*mato*
face	*yoq*
faces	*yoqlar*
factor	*ko’paytuvchi*
factored	*ko’paytuvchilarga ajratilgan*
factoring	*ko’paytuvchilarga ajratish*
factors	*ko’paytuvchilar*
fahrenheit	*farengeyt*
false	*noto’g’ri*
fee	*to’lov*
feet	*fut*
female	*ayol*
fertilizer	*o’g’it*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
fewer	*kamroq*
field	*maydon*
figure	*shakl*
filled	*to’ldirilgan*
filler	*to’ldiruvchi*
film	*film*
find	*topish*
finding	*topish*
first	*birinchi*
fish	*baliq*
fit	*mos kelish*
fixed	*o’zgarmas*
flags	*bayroqlar*
flight	*parvoz*
floor	*pol*
fluid	*suyuqlik*
fly	*uchmoq*
following	*quyidagi*
follows	*kelib chiqadi*
food	*oziq-ovqat*
foot	*fut*
forest	*o’rmon*
form	*ko’rinish*
formed	*hosil bo’lgan*
formula	*formula*

English	O’zbekcha
found	*topilgan*
fourth	*to’rtinchi*
fraction	*kasr*
free	*bepul*
frequency	*chastota*
fruit	*meva*
frying	*qovurish*
full	*to’liq*
function	*funksiya*
functions	*funksiyalar*
gallon	*gallon*
gallons	*gallonlar*
game	*o’yin*
games	*o’yinlar*
garden	*bog’*
gas	*gaz*
gasoline	*benzin*
geologist	*geolog*
geometry	*geometriya*
get	*olmoq*
gift	*sovg’a*
give	*bermoq*
given	*berilgan*
gives	*beradi*
goal	*maqsad*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
grade	*sinf*
gram	*gramm*
grams	*gramm*
graph	*grafik*
graphed	*grafikka tushirilgan*
graphs	*grafiklar*
gray	*kulrang*
greater	*katta*
greatest	*eng katta*
green	*yashil*
ground	*yer*
group	*guruh*
growth	*o’sish*
gym	*sportzal*
habitat	*yashash muhiti*
half	*yarim*
halfway	*yarim yo’lda*
hand	*qo’l*
haul	*tashimoq*
hectare	*gektar*
height	*balandlik*
heights	*balandliklar*
high	*yuqori*
highest	*eng yuqori*
histogram	*gistogramma*

English	O’zbekcha
hits	*tegadi (uriladi)*
home	*uy*
horizontal	*gorizontal*
hour	*soat*
hourly	*soatlik*
hours	*soat*
house	*uy*
houses	*uylar*
however	*ammo*
hypotenuse	*gipotenuza*
ice	*muz*
iceland	*Islandiya*
identical	*aynan bir xil*
igneous	*otqin*
inch	*dyuym*
inches	*dyuym*
include	*o’z ichiga olmoq*
included	*kiritilgan*
includes	*o’z ichiga oladi*
incorrectly	*noto’g’ri*
increase	*oshish*
increased	*ortdi*
increases	*ortadi*
increasing	*o’suvchi*
indicates	*ko’rsatadi*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
inequalities	*tengsizliklar*
inequality	*tengsizlik*
infinitely	*cheksiz*
information	*ma’lumot*
initial	*boshlang’ich*
inside	*ichida*
installation	*o’rnatish*
instead	*o’rniga*
integer	*butun son*
integers	*butun sonlar*
intercept	*kesma*
intercepts	*kesishish nuqtalari*
interior	*ichki*
intermediate	*oraliq*
internet	*internet*
internships	*stajirovka*
interpretation	*talqin*
intersect	*kesishmoq*
intersected	*kesishgan*
intersection	*kesishish*
intersects	*kesishadi*
interval	*oraliq*
isn’t	*emas*
isosceles	*teng yonli*
items	*buyumlar*

English	O’zbekcha
january	*yanvar*
jars	*bankalar*
job	*ish*
join	*qo’shilmoq*
joules	*joul*
juice	*sharbat*
july	*iyul*
kelvins	*kelvinlar*
kilograms	*kilogramm*
kilometer	*kilometr*
kilometers	*kilometr*
kinetic	*kinetik*
labeled	*belgilangan*
lake	*ko’l*
land	*yer*
large	*katta*
larger	*kattaroq*
largest	*eng katta*
last	*oxirgi*
lateral	*yon (yon sirt)*
launch	*uchirish*
launched	*uchirildi*
left	*chap*
leg	*katet*
legs	*katetlar*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
lemons	*limonlar*
length	*uzunlik*
lengths	*uzunliklar*
let	*faraz qilaylik*
level	*saviya*
lie	*yotmoq*
lies	*yotadi*
like	*o’xshash*
likely	*ehtimol*
line	*chiziq*
linear	*chiziqli*
lines	*chiziqlar*
liquid	*suyuqlik*
list	*ro’yxat*
listed	*ro’yxatda keltirilgan*
liter	*litr*
liters	*litr*
local	*mahalliy*
located	*joylashgan*
locations	*joylashuv*
long	*uzun*
longer	*uzunroq*
longest	*eng uzun*
low	*past*
lowest	*eng past*

English	O’zbekcha
machine	*mashina*
made	*qilingan*
magazines	*jurnallar*
maintain	*saqlab turish*
make	*yasamoq*
makes	*qilmoq*
making	*hosil qilish*
maple	*klyon*
marble	*sharcha*
marbles	*sharchalar*
margin	*chegara*
marked	*belgilamoq*
mass	*massa*
materials	*materiallar*
maximum	*maksimum*
mean	*o’rtacha*
means	*anglatadi*
measure	*o’lchov*
measured	*o’lchangan*
measures	*o’lchovlari*
measuring	*o’lchash*
median	*mediana*
medians	*medianalar*
meet	*uchrashmoq*
member	*a’zo*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
members	*a’zolar*
metal	*metall*
meter	*metr*
meters	*metr*
method	*usul*
middle	*o’rta*
midpoint	*o’rta nuqta*
mile	*milya*
miles	*milya*
milliliter	*millilitr*
millimeters	*millimetr*
million	*million*
minimum	*minimum*
minus	*minus*
minute	*daqiqa*
minutes	*daqiqa*
mirror	*ko’zgu*
misidentifying	*noto’g’ri aniqlash*
misinterpreting	*noto’g’ri talqin qilish*
mixture	*aralashma*
mode	*moda*
model	*model*
modeled	*modellashtirildi*
models	*modellar*
monarch	*monarx (kapalak turi)*

English	O’zbekcha
money	*pul*
month	*oy*
monthly	*oylik*
months	*oylar*
moon	*oy*
movie	*kino*
movies	*filmlar*
moving	*harakatlanayotgan*
muffins	*maffin*
multiplied	*ko’paytirildi*
multiply	*ko’paytirmoq*
multiplying	*ko’paytirish*
museum	*muzey*
nearest	*eng yaqin*
need	*kerak*
needed	*kerakli*
needs	*kerak bo’ladi*
negative	*manfiy*
never	*hech qachon*
new	*yangi*
next	*keyingi*
nonlinear	*nochiziqli*
normal	*normal*
note	*eslatma*
novels	*romanlar*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
number	*son*
numbers	*sonlar*
numerator	*surat*
object	*jism*
objects	*narsalar*
observation	*kuzatish*
obtuse	*o’tmas (burchak)*
occurs	*ro’y beradi*
ocean	*okean*
odd	*toq*
office	*ofis*
old	*eski*
onetime	*bir martalik*
online	*onlayn*
opened	*ochildi*
opens	*ochadi*
opposite	*qarama-qarshi*
oranges	*apelsinlar*
order	*tartib*
ordered	*tartiblangan*
original	*dastlabki*
ounces	*unsiya*
outcomes	*natijalar*
outfits	*kiyimlar*
output	*chiqish (natija)*

English	O’zbekcha
owner	*egasi*
packages	*paketlar*
paid	*to’lamoq*
paint	*bo’yoq*
pair	*juft*
pairs	*juftlar*
pans	*tovalar*
pants	*shim*
paper	*qog’oz*
parabola	*parabola*
parallel	*parallel*
parallelogram	*parallelogramm*
parentheses	*qavs*
park	*park*
parking	*avtoturargoh*
part	*qism*
participant	*ishtirokchi*
participants	*ishtirokchilar*
particular	*muayyan*
parts	*qismlar*
party	*ziyofat*
pass	*o’tmoq*
passes	*o’tmoq*
passing	*o’tish*
payment	*to’lov*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
pens	*ruchkalar*
people	*odamlar*
per	*har biriga*
percent	*foiz*
percentage	*foiz*
percentages	*foizlar*
perimeter	*perimetr*
perimeters	*perimetr*
period	*davr*
periods	*davrlar*
perpendicular	*perpendikulyar*
person	*odam*
phone	*telefon*
piece	*bo’lak*
pieces	*donalar*
placed	*joylashtirilgan*
plan	*reja*
plane	*tekislik*
planet	*sayyora*
plans	*rejalar*
plant	*o’simlik*
planted	*ekildi*
plants	*o’simliklar*
platform	*platforma*
plausible	*asosli*

English	O’zbekcha
play	*o’ynamoq*
played	*o’ynagan*
player	*o’yinchi*
players	*o’yinchilar*
plot	*belgilash*
plus	*plyus*
point	*nuqta*
points	*nuqtalar*
polynomial	*ko’phad*
population	*aholi*
positive	*musbat*
possible	*mumkin bo’lgan*
poster	*plakat*
posters	*plakatlar*
potatoes	*kartoshka*
pound	*funt*
pounds	*funt*
predicted	*bashorat qilingan*
predicts	*bashorat qiladi*
premium	*sug’urta badali*
prepare	*tayyorlash*
preserve	*saqlamoq*
pressure	*bosim*
previous	*oldingi*
price	*narx*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
prism	*prizma*
probability	*ehtimollik*
problem	*masala*
produce	*ishlab chiqarmoq*
produced	*ishlab chiqardi*
produces	*ishlab chiqaradi*
product	*ko’paytma*
production	*ishlab chiqarish*
profit	*foyda*
program	*dastur*
project	*loyiha*
projectile	*otilgan jism*
propellant	*yonilg’i*
properties	*xossalar*
property	*xossa*
proportion	*proportsiya*
proportional	*proporsional*
proposal	*taklif*
protein	*oqsil*
prove	*isbotlamoq*
purchase	*xarid*
purchased	*sotib olingan*
pyramid	*piramida*
pythagorean	*pifagor*
quadratic	*kvadrat*

English	O’zbekcha
quantity	*miqdor*
rabbit	*quyon*
radians	*radian*
radius	*radius*
ran	*yugurdi*
random	*tasodifiy*
randomly	*tasodifiy ravishda*
range	*diapazon*
raspberries	*malina*
rate	*tezlik*
rates	*tezliklar*
rather	*aksincha*
ratio	*nisbat*
ratios	*nisbatlar*
reach	*yetmoq*
reached	*yetdi*
reaches	*yetadi*
read	*o’qimoq*
real	*haqiqiy*
received	*oldi*
reciprocal	*teskari son/ifoda*
recorded	*qayd qilindi*
rectangle	*to’g’ri to’rtburchak*
rectangles	*to’g’ri to’rtburchaklar*
rectangular	*to’g’ri burchakli*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
red	*qizil*
reduced	*qisqartirilgan*
region	*soha*
registered	*ro’yxatdan o’tgan*
regular	*muntazam*
relates	*bog’liq*
relationship	*bog’liqlik*
relationships	*munosabatlar*
released	*chiqarilgan*
remain	*qolish*
remaining	*qolgan*
remains	*o’zgarmay qoladi*
removed	*olib tashlangan*
rent	*ijara haqi*
rental	*ijara*
renting	*ijaraga olish*
repair	*ta’mirlash*
reported	*qayd etilgan*
represent	*ifodalamoq*
represented	*ifodalangan*
representing	*ifodalovchi*
represents	*ifodalaydi*
residents	*aholi*
respectively	*mos ravishda*
responded	*javob berdi*

English	O’zbekcha
rest	*qolgan qism*
restaurant	*restoran*
restaurants	*restoranlar*
result	*natija*
resulting	*natijada hosil bo’lgan*
results	*natijalar*
retail	*chakana*
revenue	*daromad*
rewriting	*qayta yozish*
rewritten	*qayta yozilgan*
ride	*minmoq*
right	*to’g’ri*
rocket	*raketa*
rocks	*toshlar*
rolls	*nonchalar*
room	*xona*
root	*ildiz*
round	*yaxlitlamoq*
rows	*qatorlar*
running	*yugurish*
salads	*salatlar*
salary	*maosh*
sale	*sotuv*
sales	*sotuv*
salt	*tuz*

VOCABULARY – COMPLETE GLOSSARY

English	O'zbekcha
sample	*tanlanma*
samples	*namunalar*
sandwich	*sendvich*
satisfies	*qanoatlantiradi*
satisfy	*qanoatlantiradi*
save	*jamg'arish*
savings	*jamg'arma*
scale	*masshtab*
scatterplot	*sochilma diagramma*
scatterplots	*sochilish diagrammalari*
school	*maktab*
scientist	*olim*
screws	*vintlar*
scuba	*skuba*
sea	*dengiz*
seals	*tyulenlar*
seating	*o'tirish joylari*
seats	*o'rindiqlar*
second	*ikkinchi*
seconds	*soniya*
seeds	*urug'lar*
segment	*kesma*
segments	*kesmalar*
selected	*tanlangan*
selecting	*tanlash*

English	O'zbekcha
selling	*sotish*
sells	*sotmoq*
sent	*yubormoq*
september	*sentyabr*
service	*xizmat*
session	*mashg'ulot*
sessions	*mashg'ulotlar*
set	*to'plam*
sets	*to'plamlar*
setting	*o'rnatish*
shaded	*bo'yalgan*
shape	*shakl*
shifting	*siljish*
shipment	*yuk jo'natmasi*
shirt	*ko'ylak*
shirts	*ko'ylaklar*
shorter	*qisqaroq*
shortest	*eng qisqa*
show	*ko'rsatish*
shown	*ko'rsatilgan*
shows	*ko'rsatadi*
side	*tomon*
sides	*tomonlar*
sign	*ishora*
silicon	*kremniy*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
similar	*o’xshash*
similarly	*xuddi shunday*
simplifies	*soddalashtiradi*
simplifying	*soddalashtirish*
sine	*sinus*
site	*joy (maydon)*
situation	*vaziyat*
size	*o’lcham*
slope	*qiyalik*
slopes	*qiyaliklar*
small	*kichik*
smaller	*kichikroq*
smallest	*eng kichik*
snails	*shilliqqurtlar*
snow	*qor*
snowfall	*qor yog’ishi*
sold	*sotildi*
solution	*yechim*
solutions	*yechimlar*
solve	*yechmoq*
solved	*yechmoq*
solving	*yechish*
space	*bo’shliq*
species	*turlar*
speed	*tezlik*

English	O’zbekcha
spend	*sarflamoq*
spends	*sarflaydi*
spent	*sarflagan*
sphere	*shar*
spoons	*qoshiqlar*
sport	*sport*
spread	*yoyilish*
spring	*bahor*
square	*kvadrat*
squared	*kvadratga ko’tarilgan*
squares	*kvadratlar*
squaring	*kvadratga ko’tarish*
standard	*standart*
star	*yulduz*
stars	*yulduzlar*
start	*boshlanish*
started	*boshlandi*
starts	*boshlanadi*
state	*shtat*
statement	*tasdiq*
statements	*tasdiqlar*
states	*shtatlar*
station	*bekat*
statistics	*statistika*
storage	*saqlash*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
store	*do’kon*
straight	*to’g’ri*
student	*o’quvchi*
students	*o’quvchilar*
study	*tadqiqot*
subscribed	*obuna bo’ldi*
subscribers	*obunachilar*
substance	*modda*
substituted	*o’rniga qo’ymoq*
substituting	*o’rniga qo’yish*
substitution	*o’rniga qo’yish*
subtracting	*ayirish*
sufficient	*yetarli*
suits	*kostyumlar*
sum	*yig’indi*
summarizes	*umumlashtirmoq*
summer	*yoz*
supplementary	*to’ldiruvchi*
support	*qo’llab-quvvatlamoq*
surface	*sirt*
survey	*so’rovnoma*
surveyed	*so’rov o’tkazilgan*
swimmers	*suzuvchilar*
switching	*almashtirish*
symmetric	*simmetrik*

English	O’zbekcha
system	*sistema*
systems	*sistemalar*
table	*jadval*
tables	*jadvallar*
tablet	*planshet*
take	*olish*
takes	*oladi*
taking	*olish*
tall	*baland*
taller	*balandroq*
tangent	*urinma*
tank	*bak*
tape	*lenta*
tax	*soliq*
teacher	*o’qituvchi*
team	*jamoa*
teaspoons	*choy qoshiq*
teens	*o’smirlar*
televisions	*televizor*
temperature	*harorat*
temperatures	*haroratlar*
tent	*chodir*
tenth	*o’ninchi*
tents	*chodirlar*
term	*had*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
terms	*hadlar*
theater	*teatr*
theorem	*teorema*
therefore	*demak*
third	*uchinchi*
thousand	*ming*
through	*orqali*
thrown	*otilgan*
thus	*demak*
ticket	*chipta*
tickets	*chiptalar*
tiles	*plitka*
time	*vaqt*
times	*marta*
together	*birgalikda*
token	*jeton*
tokens	*jetonlar*
took	*olmoq*
total	*jami*
town	*shaharcha*
trail	*yo’lka*
trailer	*tirkama*
train	*poyezd*
training	*mashg’ulot*
translating	*tarjima qilish*

English	O’zbekcha
transversal	*kesuvchi*
trapezoid	*trapetsiya*
travel	*harakatlanish*
traveled	*bosib o’tdi*
traveling	*harakatlanmoq*
travels	*harakatlanadi*
tree	*daraxt*
trees	*daraxtlar*
triangle	*uchburchak*
triangles	*uchburchaklar*
trigonometry	*trigonometriya*
trip	*sayohat*
truck	*yuk mashinasi*
true	*to’g’ri*
twice	*ikki marta*
type	*tur*
types	*turlari*
unit	*birlik*
united	*birlashgan*
units	*birliklar*
upward	*yuqoriga*
use	*foydalanish*
used	*qo’llaniladi*
uses	*foydalanadi*
using	*yordamida*

VOCABULARY – COMPLETE GLOSSARY

English	O’zbekcha
utility	*kommunal xizmat*
valley	*vodiy*
value	*qiymat*
values	*qiymatlar*
variable	*o’zgaruvchi*
variables	*o’zgaruvchilar*
vase	*vaza*
vehicle	*transport vositasi*
vertex	*cho’qqi*
vertical	*vertikal*
video	*video*
views	*ko’rishlar*
vine	*tok*
visitors	*tashrif buyuruvchilar*
volume	*hajm*
vote	*ovoz*
voted	*ovoz bermoq*
votes	*ovozlar*
wage	*ish haqi*
walk	*yurish*
walnuts	*yong’oq*
water	*suv*
wax	*mum*
ways	*usullar*
week	*hafta*

English	O’zbekcha
weighs	*tortadi (og’irligi)*
weight	*og’irlik*
weights	*og’irliklar*
whale	*kit*
white	*oq*
whole	*butun*
width	*kenglik*
within	*ichida*
won	*yutgan*
words	*so’zlar*
work	*ish*
worked	*ishlamoq*
worker	*ishchi*
working	*ishlash*
works	*ishlaydi*
worth	*qiymat*
written	*yozilgan*
yards	*yard*
year	*yil*
years	*yillar*
yellow	*sariq*
yield	*bermoq*
yields	*beradi*
zebras	*zebralar*
zero	*nol*

Matematikadan qo’rqmang — uni tushunib yeching.

SAT 1200+ seriyasining **1-kitobi** sizni asosiy arifmetikadan boshlab, algebraik ifodalar va koordinatalar tekisligigacha bo’lgan yo’lni bosqichma-bosqich, aniq va tushunarli tarzda o’tkazadi. Har bir dars — formula sahifasi, tushuncha, lug’at va mashqlar bilan to’liq mustahkamlanadi.

- 9 ta tizimli dars, har biri o’z **formula sahifasi** bilan
- **1000+** so’zdan iborat SAT lug’ati — darsma-dars va kitob oxirida to’liq ro’yxat
- Haqiqiy SAT savollar bazasi tahliliga asoslangan mashqlar
- O’zbek tilida tushuntirish, ingliz tilida amaliyot — 2-kitobga tayyor holda o’tasiz

*SAT 1200+ seriyasi: **1-kitob** (Asoslar) → **2-kitob** (Asosiy mavzular) → **3-kitob** (Yuqori ball strategiyalari)*

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